BusinessOS AI — Master Product Specification

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1. Executive Summary

BusinessOS AI is an AI-powered business operating system built for entrepreneurs, small business owners, and MSMEs who currently have to stitch together 10–20 disconnected tools (website builders, CRMs, marketing tools, accounting software, inventory spreadsheets, social media schedulers) just to run a single small business.

BusinessOS AI replaces that fragmented toolchain with **one connected workspace** that takes a person from a raw business idea through setup, branding, product/service definition, online presence, marketing, sales, customer management, operations, and growth — with AI agents doing as much of the manual work as possible.

The **business owner is the primary user.** The product is not a marketplace, not an Amazon clone, and not a generic storefront builder. Those are downstream capabilities that emerge naturally once a business is operating inside BusinessOS — they are explicitly **Phase 7+** concerns, not MVP concerns.

This document defines the full product surface, the AI agent system, the data model, the technical architecture, the relationship to the existing SMART-BUSSINESS/AgentHub AI repository, the MVP, and a phased roadmap. It intentionally avoids any implementation or coding decisions — its purpose is to make the product completely legible before a single line of Milestone 1 code is written.

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2. Product Vision

Vision statement:

> *"BusinessOS AI helps turn a business idea into a functioning, growing business — with an AI system that plans, builds, and operates alongside the owner."*

A first-time founder should be able to arrive with nothing but an idea — *"I want to start a clothing business"* — and be guided, with AI assistance at every step, through:

Idea → Research → Business Planning → Brand Creation → Product/Service Setup → Website/Store → Marketing → Customer Acquisition → Sales → Operations → Analytics → Growth

The product should feel like a **single AI-native operating system for a business**, not a bundle of SaaS tools wearing the same login page. Every screen, module, and agent exists to answer one of three questions for the owner at any moment:

- What is happening in my business right now?
- What should I do next?
- What can AI do for me automatically?

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3. Problem Statement

Small business owners and first-time entrepreneurs face a consistent set of structural problems:

Problem	Description
Tool fragmentation	Website builder + CRM + accounting + social scheduler + inventory sheet + email tool, none of which share data
No starting point	People with an idea don't know the sequence of steps to actually launch
Manual overhead	Product descriptions, marketing copy, social posts, and admin tasks are all done by hand
Weak marketing skill	Most small owners are not trained marketers or copywriters
Low technical skill	No ability to build/maintain a website or configure integrations
Operational blind spots	No unified view of inventory, orders, customers, and cash in one place
No forward guidance	Existing tools report what happened; they rarely say what to do next
Cost sensitivity	Cannot justify 6–10 separate SaaS subscriptions on thin margins
Automation illiteracy	Owners don't know what *can* be automated, let alone how

BusinessOS AI's thesis is that **AI agents plus a unified data model** can collapse this fragmented toolchain into a single connected system that actively reduces the owner's manual workload rather than just presenting more dashboards.

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4. Target Users

4.1 Primary Persona Segments

Persona	Description	Core Need
First-time founder	Has an idea, no business experience	Guided setup, simplified decisions
Solo entrepreneur / freelancer	Runs a one-person service or product business	Time savings, automation
MSME owner	Small registered business, some existing operations	Digitization, centralization
Online seller	Sells via Instagram/WhatsApp/marketplaces already	Structure, inventory, CRM
Local business owner	Physical shop, service provider	Simple digital presence, customer tracking
Digitizing existing business	Has manual records (Excel, paper)	Migration into structured system

4.2 Common Pain Points

- Doesn’t know where to start
- Too many disconnected tools, no shared data
- Manual, repetitive admin work
- Weak or no marketing skills
- Limited technical/design skills
- Inventory and customer records kept informally or not at all
- No financial visibility beyond a bank balance
- No analytics or feedback loop on what’s working
- Doesn’t know what could be automated
- Doesn’t know what to do next at any given stage

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5. Product Principles

1. **One central business workspace** — a single place, not a suite of separately-branded tools.
2. **AI-first** — AI is a first-class participant in the workflow, not a bolt-on chat widget.
3. **Automation-first** — repetitive tasks should default to automated, with human approval where risk exists.
4. **Beginner-friendly** — a user with zero business or technical background should be able to progress.
5. **Modular architecture** — modules can be built, shipped, and even disabled independently.
6. **Data connected across modules** — a product, order, customer, and campaign all reference the same underlying records.
7. **Owner remains in control** — AI recommends and drafts; irreversible or external-facing actions require approval by default.
8. **Reduce repetitive manual work** — every module should measurably remove a task from the owner’s plate.
9. **Start small, architect for scale** — MVP is deliberately narrow; the data model and services are not.
10. **Every feature serves the journey** — starting, operating, or growing a business — nothing else earns a place in the product.

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6. Core User Journey

Step 1 — Sign up. Owner creates an account.

Step 2 — Idea capture. BusinessOS asks: *“What are you trying to build?”*

Step 3 — Idea description. Owner describes it in plain language (e.g., “I want to start a women’s clothing business”).

Step 4 — AI Business Setup Agent interprets the idea: category, business model, likely target customer, initial structure.

Step 5 — Workspace creation. BusinessOS instantiates an initial Business + BusinessProfile.

Step 6 — Guided setup. AI assists with: business model definition, target customer, lightweight market/competitor research, positioning, brand name and identity direction, initial product/service structure, pricing suggestions, and an initial marketing strategy.

Step 7 — Product/service creation. Owner adds their first products or services.

Step 8 — AI content generation. BusinessOS generates product descriptions, marketing copy, social captions, campaign ideas, and (where supported) images.

Step 9 — Launch. Owner activates their business presence (initially: profile + basic public page; later: full storefront).

Step 10 — Daily operating system. BusinessOS becomes the day-to-day home base. The dashboard continuously answers: what's happening, what needs attention, what to do next, and what AI can take off the owner's plate.

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7. Product Architecture Overview

Conceptually, BusinessOS is layered:

```
Frontend (Owner-facing App)
        |
API / Application Services Layer
        |
Core Business Services (Products, CRM, Orders, Inventory, Finance, Marketing)
        |
AI Agent Layer (reasoning, tool-calling, retrieval)
        |
Automation / Workflow Engine
        |
Database (multi-tenant, per-Business isolation)
        |
Integrations (payments, social, messaging, storage)
```

Each module is a bounded service area with its own data, but all reference a shared `Business` entity so that AI agents and dashboards can reason across modules (e.g., a marketing agent can see product and order data; a finance view can see order revenue).

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8. Module Specifications

Modules are deliberately **not** all scheduled for V1. Each is tagged with its target phase.

Module	Purpose	Phase
A. Business Setup	Idea capture, AI-assisted structuring of a new business	MVP
B. Business Profile	Core business record: name, category, description, branding	MVP
C. Business Planning	Lightweight plan: goals, target customer, positioning	MVP
D. AI Business Advisor	Ongoing guidance agent, "what should I do next"	MVP (basic)
E. Product/Service Management	Create, edit, describe products/services	MVP
F. Website/Store Builder	Public-facing page/storefront	Phase 7
G. CRM	Customer records, leads, contact history	MVP (basic)

H. Sales	Sales recording, pipeline for service businesses	MVP (basic)
I. Orders	Order capture and lifecycle	MVP (basic)
J. Inventory	Stock tracking, low-stock alerts	Phase 6
K. Marketing	Campaign planning, AI copy generation	Phase 5 (basic in MVP)
L. Social Media	Scheduling, publishing, content calendar	Phase 5
M. Finance	Invoicing, transactions, basic bookkeeping	Phase 6
N. Analytics	Cross-module reporting and insights	Phase 2–3 (basic in MVP)
O. Automation	Rule-based and AI-triggered workflows	Phase 8 (simple triggers earlier)
P. AI Agents	Full agent framework (see Section 9)	Phased, starting MVP
Q. Notifications	Alerts, reminders, digest emails	MVP (basic)
R. Integrations	Payments, social platforms, messaging	Phased by need
S. Customer Support	Owner’s support inbox / AI-assisted replies	Phase 5–6
T. Team Management	Multi-user roles per business	Phase 4+

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9. AI Agent Architecture

AI agents in BusinessOS are **not chatbots**. Each agent is a reasoning system with defined inputs, tool access, data scope, and an explicit boundary between actions it can take autonomously and actions that require owner approval.

9.1 Agent Roster

#	Agent	Purpose
1	Business Advisor Agent	Ongoing “what should I do next” guidance across the whole business
2	Business Setup Agent	Interprets a raw idea into an initial business structure
3	Market Research Agent	Summarizes market/category context and audience signals
4	Product Research Agent	Suggests product/service structuring, pricing bands
5	Product Content Agent	Generates descriptions, specs, image prompts
6	Marketing Agent	Builds campaign concepts and copy
7	Social Media Agent	Drafts and schedules social content
8	Sales Agent	Surfaces follow-ups, drafts outreach
9	CRM Agent	Enriches/deduplicates customer records, flags at-risk customers
10	Inventory Agent	Flags stock issues, suggests reorder points
11	Finance Assistant Agent	Drafts invoices, flags anomalies, summarizes cash position
12	Customer Support Agent	Drafts replies to inbound enquiries
13	Analytics Agent	Turns raw metrics into plain-language insight

14	Automation Agent	Proposes and configures automation rules
15	Growth Advisor Agent	Longer-horizon strategic suggestions

9.2 Agent Definition Template (applies to all agents)

For every agent, the following must be defined before implementation:

- **Purpose** — the single job the agent exists to do
- **Inputs** — what data/context it receives (structured records + user prompt)
- **Outputs** — draft content, recommendation, structured action, or report
- **Data access** — which entities/tables it can read
- **Actions it can perform** — what it's allowed to write/trigger
- **Human approval requirement** — autonomous vs. approval-gated
- **Automation opportunities** — what could later run on a schedule/trigger
- **Example interaction** — a concrete sample exchange

9.3 Example: Product Content Agent (fully specified)

- **Purpose:** Generate marketable product descriptions and content drafts from raw product input.
- **Inputs:** Product name, category, attributes, target customer (from BusinessProfile), tone preference.
- **Outputs:** Draft description(s), bullet feature list, suggested title variants, image prompt (if image generation is enabled).
- **Data access:** Read — Product, BusinessProfile. Write — draft fields on Product only.
- **Actions it can perform:** Populate draft fields; it cannot publish without owner confirmation.
- **Human approval:** Required before content is published to any public-facing surface.
- **Automation opportunity:** Auto-draft on product creation; auto-regenerate on request.
- **Example interaction:** Owner adds "Cotton Kurta, Blue, ₹899" → agent returns a description, 3 title options, and 5 bullet points → owner edits and approves.

9.4 Approval Model

All agents fall into one of three tiers:

Tier	Description	Example
Autonomous	Runs without confirmation, internal-only effect	Drafting content, generating analytics summaries
Approval-gated	Drafts an action; owner must confirm before it takes effect	Publishing content, sending customer emails, posting to social
Restricted	Never fully autonomous, always requires explicit per-instance approval	Payments-related actions, refunds, data deletion

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10. BusinessOS Dashboard

The dashboard is the daily home screen and must answer, at a glance:

- How is my business doing?
- What happened today?
- What needs my attention?
- What opportunities exist?
- What should I do next?
- What can AI automate for me?

10.1 Dashboard Sections

Section	Content
Business Health Summary	Composite status: revenue trend, order volume, customer growth
Today’s Activity	New orders, new leads, messages, low-stock flags
AI Recommendations	Ranked list of suggested next actions from the Advisor Agent
Tasks	Owner and AI-generated task list
Revenue & Orders	Rolling revenue chart, order count, average order value
Customers	New vs. returning, at-risk customers flagged by CRM Agent
Inventory Alerts	Low stock, out-of-stock items (once Inventory module is active)
Marketing Performance	Campaign/content performance summary (once Marketing module is active)
AI Actions Log	What AI did automatically, and what’s pending approval

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11. Website / Storefront (Secondary Layer)

Once a business is structured inside BusinessOS, the owner should eventually be able to:

- Choose a template
- Apply branding
- Add products/services
- Publish a public page/storefront
- Receive enquiries and orders
- Have those orders and customers flow directly into BusinessOS

This capability is explicitly **secondary** to the core BusinessOS product and is scheduled for **Phase 7**. The MVP focuses entirely on the owner’s operating experience, not the public-facing storefront.

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12. Customer-Side Future Extension

Stage 1 (Phase 7+): Each BusinessOS business gets its own individual storefront. Customers browse and buy from that single business; orders flow into that business’s BusinessOS workspace.

Stage 2 (Phase 10, long-term): A BusinessOS Marketplace aggregates multiple businesses so customers can discover and purchase across sellers, while BusinessOS continues to power the operational backend for every participating business.

This sequencing is intentional: BusinessOS must first prove it can run a single business well before it aggregates many businesses into a marketplace.

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13. Database Architecture (Conceptual)

13.1 Core Entities

Entity	Purpose
User	Login identity
Organization	Top-level tenant (may map 1:1 with Business initially)
Business	The business being operated

Business	The business being operated
BusinessProfile	Branding, category, description, positioning
Product / ProductVariant	Items or services sold
Customer	End customer of the business
Lead	Pre-customer contact
Order / OrderItem	Sales transactions
Inventory	Stock levels tied to ProductVariant
Supplier	Source of inventory (Phase 6+)
Invoice / Transaction	Financial records
MarketingCampaign	Campaign definitions and content
SocialAccount / SocialPost	Connected social channels and scheduled posts
Task	Owner or AI-generated action items
Automation	Configured trigger→action rules
AIConversation	Chat/interaction history with agents
AIAgentRun	Structured log of an agent execution (inputs, outputs, approval state)
Notification	In-app/email alerts
Website / Storefront	Public-facing presence (Phase 7+)
Subscription	Billing plan state
TeamMember	Additional users on a Business (Phase 4+)

13.2 Relationship Notes

- `Organization` 1—many `Business` (supports a user eventually running multiple businesses)
- `Business` 1—many `Product`, `Customer`, `Order`, `Task`, `AIAgentRun`, etc. — this is the primary tenancy boundary
- `Order` many—many `Product` via `OrderItem`
- `Product` 1—many `ProductVariant`; `Inventory` keys off `ProductVariant`
- `AIAgentRun` references the entity it acted on (polymorphic reference) plus the agent type and approval status
- `TeamMember` links `User` to `Business` with a role

This model is deliberately conservative for MVP (fewer entities active) but designed so later phases (Inventory, Finance, Automation, Storefront) attach cleanly without re-architecting core tables.

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14. Technical Architecture

14.1 Layers

```
Frontend (React + TypeScript, Vite or Next.js)
        |  REST/RPC
API Layer (FastAPI or Node.js — single choice, TBD at Milestone 1)
        |
Business Services (Products, CRM, Orders, Inventory, Finance, Marketing)
        |
AI Agent Layer (LLM provider + tool-calling + retrieval where needed)
        |
Automation Layer (internal workflow engine; n8n only if justified later)
        |
Database (PostgreSQL, via Supabase or equivalent managed Postgres)
        |
```

```
Integrations (Payments: Razorpay/Stripe; Social APIs; Object storage)
```

14.2 Notes on Each Layer

- **Frontend:** React + TypeScript. Vite for a fast MVP; Next.js only if SSR/SEO becomes a near-term requirement (relevant once storefronts exist).
- **Auth:** Supabase Auth or equivalent — email/password + OAuth, org/business-scoped sessions.
- **Storage:** Supabase Storage or equivalent object storage for product images, generated assets.
- **AI layer:** LLM provider(s) behind an internal agent abstraction; tool-calling for structured actions (create product, draft campaign, etc.); retrieval (RAG) only where business-specific context genuinely requires it (avoid over-engineering this in MVP).
- **Automation:** Start with simple internal trigger→action rules tied to core events (new order, new customer, low stock). Introduce a dedicated workflow engine only once automation needs outgrow simple internal rules.
- **Payments:** Region-dependent — Razorpay for India-first launch, Stripe for broader/global expansion.

This is a recommended reference architecture, not a final lock-in — Milestone 1 should confirm backend language choice based on what's reusable from SMART-BUSSINESS.

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15. Existing Repository Analysis (SMART-BUSSINESS / AgentHub AI)

Important caveat: This analysis is based on the description provided (React/Vite/Supabase seller/e-commerce product with authentication, seller dashboard, product management, orders, analytics, AI content generation, image tools, marketing, social media, storefront, and Supabase Edge Functions). No direct repository access was available at the time of writing this document — before Milestone 1 begins, the actual codebase should be reviewed against this mapping to confirm or correct these assumptions.

Existing Capability (AgentHub AI)	BusinessOS Mapping	Recommendation
Authentication	Core BusinessOS auth/tenancy	Reuse, extend to Organization/Business model
Seller dashboard	BusinessOS Dashboard (Section 10)	Refactor: reframe around "what's happening / what's next," not just seller stats
Product management	Product/Service Management module	Reuse core CRUD, extend schema for services (not just physical products)
Orders	Orders module	Reuse as a starting point; simplify for MVP, extend later
Analytics	Analytics module	Reuse where metrics already exist; reframe as insight-driven, not raw charts
AI content generation	Product Content Agent / Marketing Agent	Reuse generation logic; re-architect into the formal agent framework (Section 9)
Image tools	Product Content Agent (image prompt/generation path)	Reuse if functional; treat as an agent capability, not a standalone tool
Marketing tools	Marketing module	Reuse content generation; defer campaign scheduling to Phase 5
Social media	Social Media module	Defer to Phase 5 — do not prioritize in MVP even if code exists
Storefront	Website/Storefront (Section 11)	Explicitly deprioritize to Phase 7 — do not let existing storefront code pull focus into MVP
Supabase / Edge Functions	Core infrastructure	Reuse as the default backend/database choice unless Milestone 1 review finds a blocker

What should likely be removed or set aside: any marketplace-style, multi-seller, or public-discovery features from AgentHub AI — these map to Phase 10 (Marketplace), not MVP, and should not shape early schema decisions.

What is missing and must be newly built: the Business Setup Agent and idea-to-plan flow (Section 6, Steps 2–6), the CRM module, the unified AI Agent framework with approval tiers (Section 9), and the cross-module dashboard (Section 10) as a “what’s next” surface rather than a stats page.

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16. UX Architecture — Screen List

16.1 Public

- Landing page
- Pricing page
- Login
- Signup

16.2 Onboarding

- Business idea capture
- AI-assisted business setup review
- Business profile confirmation
- Initial AI recommendations review

16.3 Core Application

Screen	Purpose	Key Components
Dashboard	Daily home view	Health summary, AI recommendations, tasks, alerts
Business	Profile, branding, positioning	Editable profile fields, brand assets
Products	Manage products/services	List, create/edit form, AI content panel
Customers	CRM	Customer list, detail view, notes/history
Sales / Orders	Transaction management	Order list, order detail, status pipeline
Inventory (Phase 6)	Stock tracking	Stock levels, low-stock alerts
Marketing (Phase 5)	Campaigns and content	Campaign list, AI content generator
Finance (Phase 6)	Invoicing, transactions	Invoice list, transaction ledger
Analytics	Cross-module insight	Charts + plain-language AI summaries
AI Agents	Agent interaction hub	Per-agent chat/action history
Automations (Phase 8)	Configured rules	Rule list, trigger/action builder
Settings	Account, team, billing	Profile, plan, team members, integrations

16.4 Future

- Website/store builder (Phase 7)
- Public storefront (Phase 7)
- Marketplace (Phase 10)

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17. Business Workflows

New Business Workflow: Account created → Business created → Idea described → AI analyzes idea → Business profile drafted → Target audience suggested → Products/services suggested → Initial action plan generated → Owner approves → Workspace activated.

Product Workflow: Product created → Details entered → AI generates content → Product saved → Inventory record created (Phase 6) → Product becomes sellable.

Sales Workflow: Order/customer event occurs → Order created → Inventory updated (Phase 6) → Customer profile updated → Revenue recorded → Dashboard updated → Analytics Agent processes transaction.

Marketing Workflow: Product selected → Marketing Agent generates campaign concept → Owner reviews → Content finalized → Scheduled/published (Phase 5) → Performance data collected → Growth/Marketing agent recommends adjustments.

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18. Automation System

Trigger	Action	Type
New order	Update inventory	Automatic
New customer	Create CRM record	Automatic
Low inventory	Alert owner	Automatic (notification)
New product	Draft marketing content	Automatic draft / Approval-gated publish
Sales milestone reached	Notify owner	Automatic
Customer inactive N days	Suggest follow-up	Approval-gated (owner sends)
Daily summary	AI-generated report	Automatic
Weekly review	AI recommendations	Automatic

General rule: **internal, reversible, informational actions are automatic; anything customer-facing, financial, or irreversible requires explicit approval**, consistent with the agent tiering in Section 9.4.

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19. MVP Definition

19.1 MVP Must Allow a User To:

1. Sign up
2. Create a business
3. Describe their business idea
4. Receive AI-assisted business setup (profile, target customer, initial positioning)
5. Create products/services (with AI-generated descriptions)
6. Manage basic customer records (CRM-lite)
7. Record basic sales/orders manually
8. View a simple, insight-oriented dashboard
9. Generate basic marketing content (copy only, no scheduling/publishing)
10. Interact with at least one useful AI agent (Business Advisor + Product Content Agent)
11. Track basic business metrics (revenue, order count, customer count)

19.2 Why Each Is Necessary

Each MVP item maps directly to removing a specific manual step identified in Section 3 (idea structuring, product copywriting, basic record-keeping) without requiring the operational depth (inventory, finance, automation, storefront) that a first-time user does not yet need to prove the core value proposition.

19.3 Explicitly NOT in MVP

- Website/storefront builder
- Public marketplace
- Inventory management
- Finance/invoicing/bookkeeping
- Social media scheduling/publishing
- Automation engine (beyond a couple of hardcoded triggers, e.g., low-stock-style placeholder)
- Team/multi-user roles
- Advanced AI agents (Market Research, Growth Advisor, Automation Agent)
- Payment processing

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20. Phased Roadmap

Phase	Focus	Key Deliverables
0	Product foundation	Finalized spec (this document), architecture decision, repo audit
1	Identity + auth	Signup/login, Organization/Business/User model, tenancy boundary
2	Business setup + AI advisor	Idea capture flow, Business Setup Agent, Business Advisor Agent (basic), Dashboard v1
3	Products/services	Product module, Product Content Agent, product list/detail UX
4	CRM + sales	Customer records, manual order entry, Sales Agent (basic), team roles begin
5	Marketing + AI agents	Marketing Agent, Social Media module, expanded agent roster
6	Inventory + finance	Inventory module, Finance module, Finance Assistant Agent
7	Website/storefront	Template builder, public page, order intake from storefront
8	Automation	Automation module, Automation Agent, rule builder UX
9	Advanced AI agents	Analytics Agent, Growth Advisor Agent, cross-agent orchestration
10	Marketplace/ecosystem	Multi-business discovery layer, shared customer marketplace

For each phase, the following should be defined at planning time (deliberately not filled in here to avoid premature technical commitment): objective, features, user value, database changes, backend work, frontend work, AI work, testing plan, and definition of done.

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21. Security & Multi-Tenancy

- **Authentication:** managed auth provider (e.g., Supabase Auth), session-scoped to Organization/Business.
- **Tenant isolation:** every core table scoped by `business_id`; enforced at the database layer (row-level security) wherever the underlying database supports it.
- **Role-based access:** Owner, Team Member (Phase 4+), with granular permissions introduced only as Team

Management is built.

- **API security:** authenticated, business-scoped endpoints; no cross-tenant data access paths.
- **Secrets management:** environment-based secrets store; no client-side exposure of API keys.
- **Audit logs:** every AI agent action logged via `AIAgentRun` with inputs, outputs, and approval state (Section 9.2).
- **AI permissions:** enforced per the three-tier approval model (Section 9.4) — autonomous, approval-gated, restricted.
- **Human approval for sensitive actions:** required for anything financial, customer-facing, or irreversible.

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22. Monetization

Illustrative tiering — final numbers require market validation, not invented here without caveat:

Tier	Target User	Illustrative Limits
Free	Trying the product	1 business, limited AI actions/month, no storefront
Starter	Early solo owner	1 business, moderate AI usage, basic marketing content
Growth	Active small business	Multiple products, CRM, basic automation, higher AI usage
Pro	Established MSME	Inventory, finance, storefront, team members
Business	Multi-user, higher volume	Full automation, advanced agents, priority support

Dimensions to gate by: number of businesses, AI actions/month, number of products, number of team members, automation rule count, storefront availability, and advanced agent access. Actual price points should be set after MVP usage data and competitor benchmarking, not fixed in this document.

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23. Competitive Positioning

Competitor	Their Focus	BusinessOS Difference
Shopify	E-commerce storefront	BusinessOS starts before the storefront — idea, planning, brand — storefront is a later layer
Zoho / Odoo	Broad business software suites	BusinessOS is AI-native and journey-driven, not a menu of disconnected apps under one login
HubSpot	CRM/marketing suite	BusinessOS integrates CRM into the full business journey, not as a standalone specialty tool
Wix	Website builder	Website is secondary in BusinessOS; the core product is business operations
QuickBooks	Accounting	Finance is one module among many, connected to sales/marketing data, not a silo
Canva	Content creation	Content generation is agent-driven and data-connected (knows the product/customer), not generic design
Zapier	Automation	Automation is embedded and business-context-aware, not a general-purpose integration tool

Traditional ERP	Large-org operations	BusinessOS is built for a solo/small owner from day one, not scaled-down enterprise software

Core differentiation: BusinessOS AI is organized around the *business journey* (idea → operating business) with AI-assisted operations throughout, rather than being one specialized tool that the owner must supplement with others.

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24. Success Metrics

North Star: *"Help a business owner go from idea to operating business with dramatically less manual work."*

Metric	What It Tells Us
Businesses created	Top-of-funnel interest
Businesses activated (first product + first sale/order logged)	Real activation, not just signup
Products/services created	Depth of setup
Orders processed	Operational usage
AI actions completed	AI adoption and reliance
Automations executed (Phase 8+)	Automation value realized
Monthly active businesses	Retention of the core unit
Customer retention (owner's customers, via CRM)	Downstream business health
Time saved per business owner (self-reported or estimated)	Core value proposition validation

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25. Risks

Risk	Mitigation
Scope creep toward marketplace/storefront too early	Enforce phase gating (Section 20); MVP explicitly excludes storefront
AI agents taking unsafe autonomous actions	Strict approval tiering (Section 9.4) applied to every agent from day one
Reusing SMART-BUSSINESS architecture uncritically	Formal repo audit against this spec before Milestone 1 (Section 15)
Over-engineering automation/agent framework before proving core value	MVP uses only Business Advisor + Product Content agents; expand roster per phase
Data model rigidity	Conservative MVP entity set (Section 13) designed to extend without core rework
Owner distrust of AI-generated content/actions	Default approval-gated tier for anything customer-facing or financial

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26. Product Boundaries — Is / Is Not

BusinessOS AI IS: - An AI-powered business operating system - A business launch assistant - A business management platform - An automation platform (progressively) - An AI business assistant - Future commerce/store infrastructure (later phase)

BusinessOS AI IS NOT, initially: - An Amazon clone - A public marketplace - A pure website builder - A pure CRM - Pure accounting software - A pure marketing tool - A generic AI chatbot

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27. CTO Final Recommendation

1. **Build first:** Identity/tenancy (Phase 1) → Business Setup + AI Advisor (Phase 2) → Products + Content Agent (Phase 3) → basic CRM/Sales (Phase 4). This is the MVP as defined in Section 19.
2. **Do not build yet:** Storefront, marketplace, inventory, finance, automation engine, social scheduling, advanced agents. All deferred per the phased roadmap (Section 20).
3. **Reuse from SMART-BUSSINESS:** Authentication foundation, core product CRUD, existing AI content-generation logic (re-homed into the formal agent framework), and Supabase infrastructure — pending a direct repository audit to confirm code quality and fit.
4. **Architecture:** React/TypeScript frontend (Vite initially), Postgres via Supabase (or equivalent), a single backend language decided at Milestone 1 based on what's reusable, and an internal AI agent abstraction layer from day one rather than ad hoc LLM calls scattered across features.
5. **First real MVP:** Idea capture → AI-assisted business setup → product creation with AI content → basic CRM/orders → insight-driven dashboard, exactly as scoped in Section 19.
6. **Milestone 1:** Identity, tenancy, and the Business/BusinessProfile data model — the foundation everything else depends on.
7. **Milestone 2:** Business Setup Agent + idea-to-plan flow + Business Advisor Agent v1 + Dashboard v1.
8. **Milestone 3:** Product/Service module + Product Content Agent + basic CRM and manual order entry.
9. **AI agent introduction:** Start with exactly two agents (Business Setup, Product Content) inside the formal approval-tiered framework from Section 9 — resist adding more agents until the core loop (idea → product → sale) is proven with real users.
10. **Customer storefront:** Introduce at Phase 7, only after Products, CRM, Sales, Marketing, and Inventory/Finance are stable — the storefront should sit on top of a proven operational core, not precede it.
11. **Marketplace:** Introduce at Phase 10, only after multiple individual storefronts are live and generating real order volume — a marketplace without operationally healthy individual businesses underneath it has nothing to aggregate.

PART 2 — VISUAL ARCHITECTURE & IMPLEMENTATION BLUEPRINT

Part 1 defined *what* BusinessOS AI is. Part 2 makes the system's structure, data flow, and interfaces visually explicit — for founders, PMs, engineers, AI engineers, designers, DevOps, and investors alike — and closes with an exact, milestone-by-milestone build order.

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26. Visual Product Architecture — Overview

Every diagram below follows the same conventions:

- **Solid arrows** = direct calls / data flow
- **Grouped boxes** = a logical layer or service boundary
- **Labeled edges** = what data or action moves along that path
- Frontend, backend, AI, and external-integration boundaries are always visually separated

These diagrams are the shared mental model for the whole team — a PM should be able to point at a box and know what it means, and an engineer should be able to point at the same box and know what to build.

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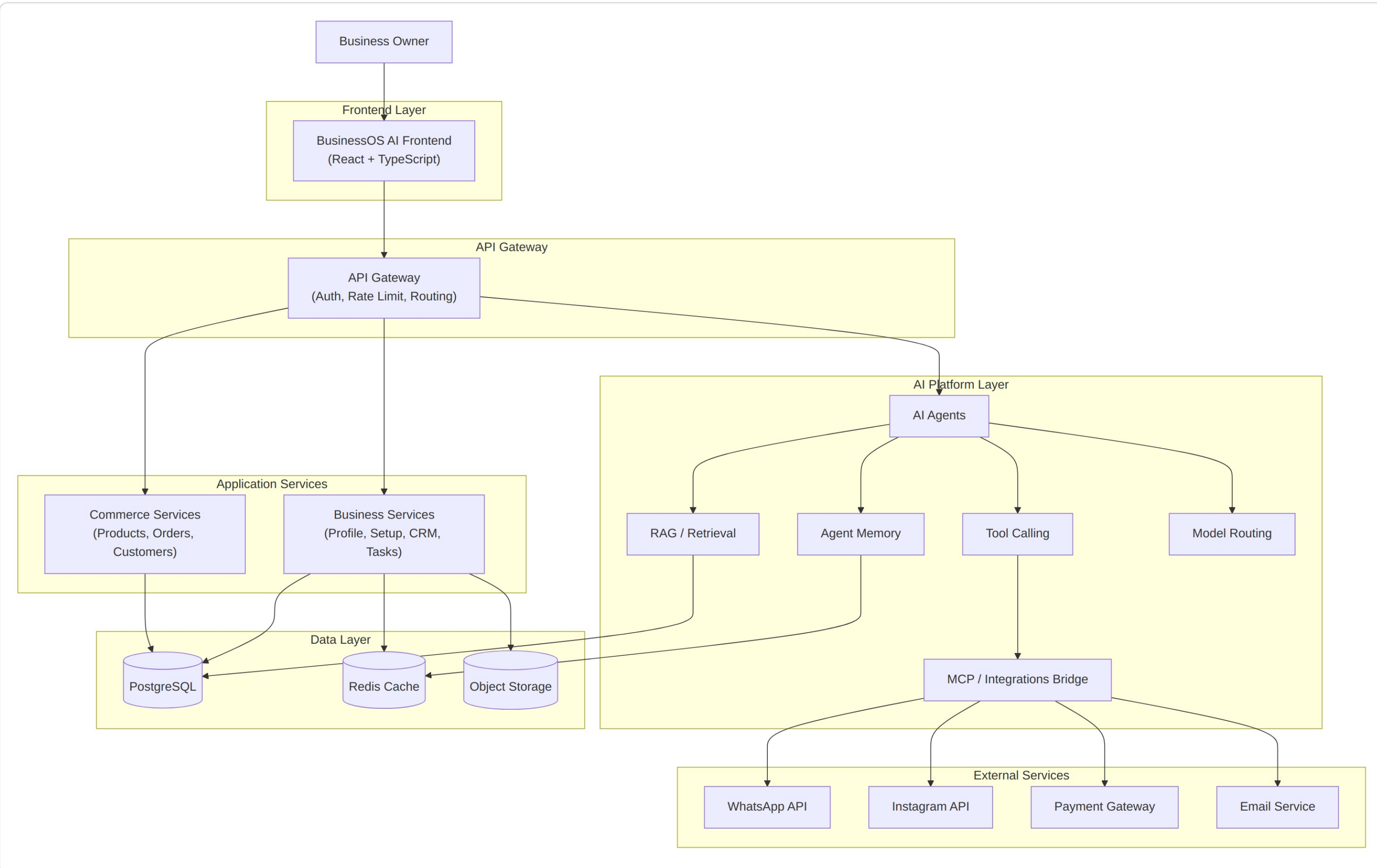
27. Master BusinessOS Architecture



Layer explanations:

- **Frontend Layer** — the owner-facing React application; the only layer the user directly touches.
- **API Gateway** — single entry point handling authentication, authorization, rate limiting, and routing to the correct

service.

- **Application Services** — the business logic: Business Services (setup, profile, CRM, tasks) and Commerce Services (products, orders, customers). This is where MVP work concentrates.
- **AI Platform Layer** — agents, tool-calling, retrieval, memory, model routing, and the MCP bridge that lets agents call out to integrations safely.
- **Data Layer** — PostgreSQL as the system of record, Redis for cache/session/short-term agent memory, object storage for images and generated assets.
- **External Services** — WhatsApp/Instagram (Phase 5+), payment gateway (Phase 6+), email (MVP, for notifications).

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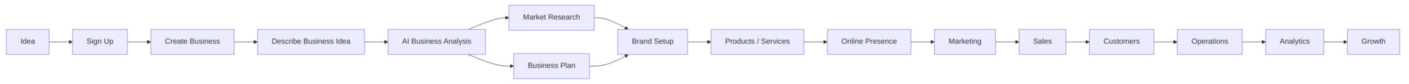
28. Business Owner Journey Diagram



What BusinessOS does at each stage:

Stage	System Behavior
Idea	Captures free-text description
Sign Up	Creates User + Organization
Create Business	Creates Business record, empty BusinessProfile
Describe Idea	Business Setup Agent parses category, model, audience signals
AI Business Analysis	Produces structured draft: profile, positioning, initial plan
Market Research	Market Research Agent adds category/competitor context (Phase 2+)
Business Plan	Owner reviews/edits AI-drafted plan and target customer
Brand Setup	Name, logo/asset placeholders, tone of voice captured in BusinessProfile
Products/Services	Owner adds items; Product Content Agent drafts descriptions
Online Presence	Basic public profile page (MVP); full storefront (Phase 7)
Marketing	AI-generated copy (MVP); scheduling/publishing (Phase 5)
Sales	Manual order entry (MVP); pipeline tracking (Phase 4+)
Customers	CRM record creation and history tracking
Operations	Inventory, finance visibility (Phase 6)
Analytics	Dashboard metrics (MVP); AI-generated insight narratives (Phase 3+)
Growth	Growth Advisor Agent surfaces longer-horizon suggestions (Phase 9)

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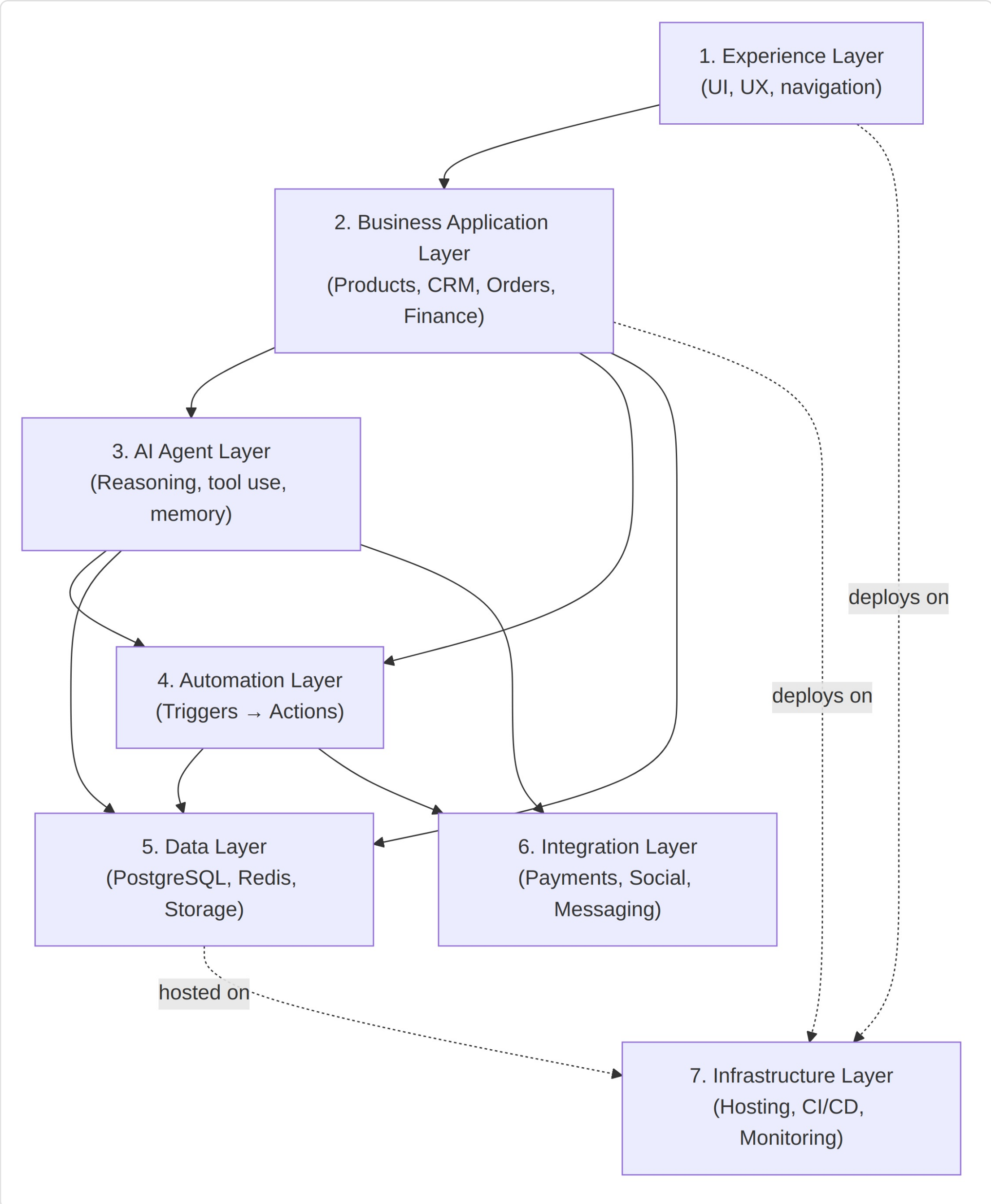
29. BusinessOS Core Architecture — Seven Layers



Communication rules:

- The Experience Layer never talks to Data or Integration layers directly — everything routes through the Business Application Layer or the AI Agent Layer.
- The AI Agent Layer can read Data Layer records (via services, not direct DB access) and can trigger the Automation Layer.
- The Automation Layer is the only layer permitted to call external Integrations autonomously (under the approval rules in Section 9.4); the AI Agent Layer calls integrations only through the Automation/MCP bridge.
- The Infrastructure Layer underlies everything but has no product logic of its own.

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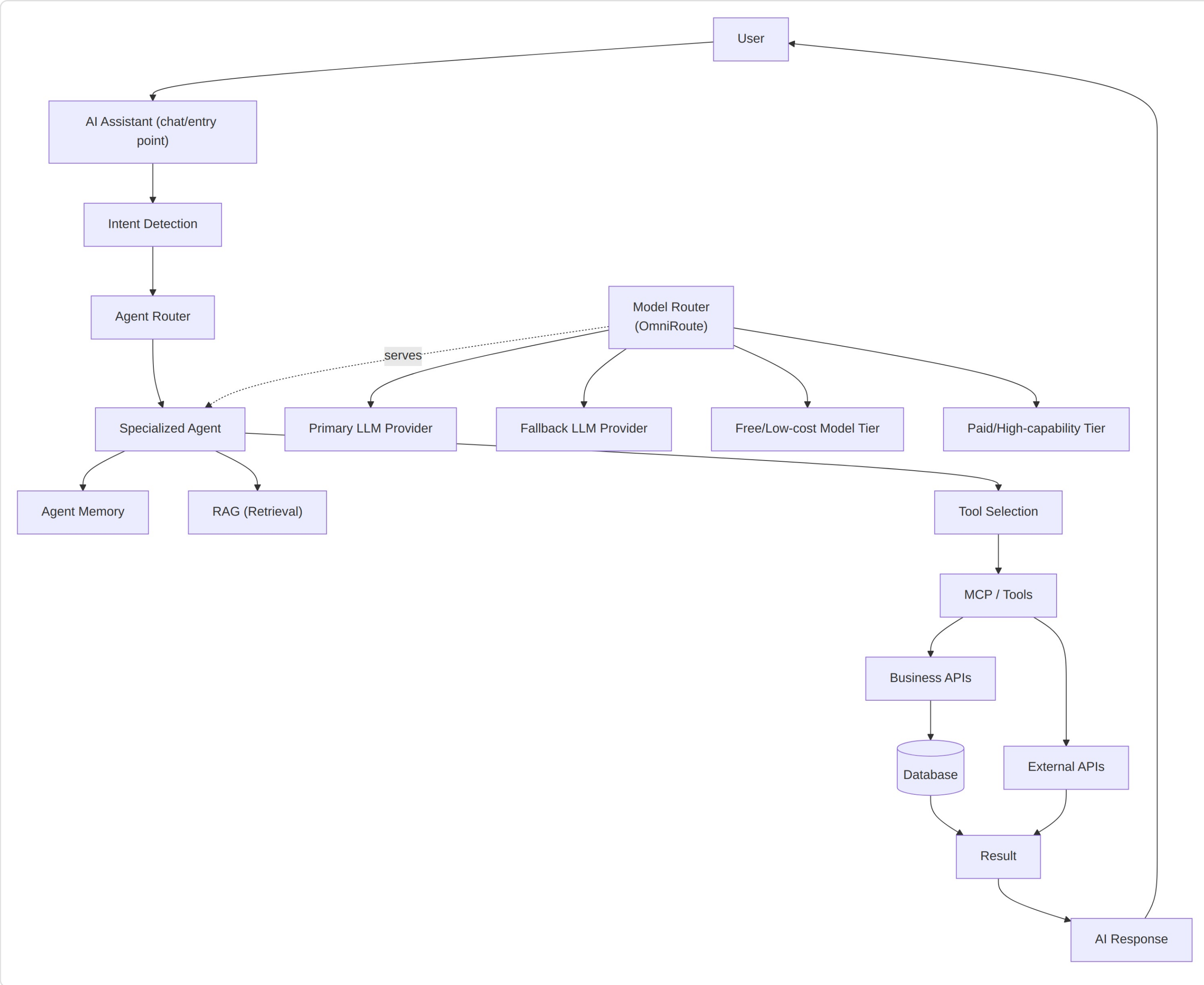
30. AI Architecture Diagram



30.1 MVP AI Architecture

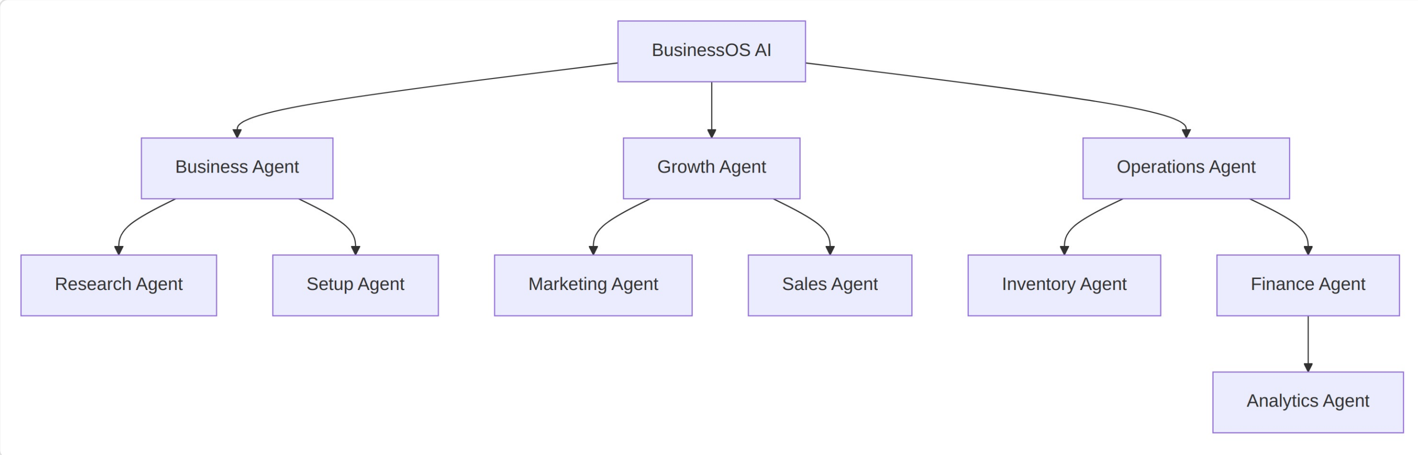
- User → AI Assistant → **single-purpose call** to one of two agents (Business Setup Agent, Product Content Agent)
- No agent router needed yet — the UI context (which screen the user is on) determines which agent is invoked
- No persistent long-term memory — conversation-scoped context only
- Single LLM provider, no fallback routing
- No RAG — all context passed directly as structured data (Business, Product records)

30.2 Future AI Architecture (Phase 5+)

- Full **Intent Detection + Agent Router** across the expanded agent roster (Section 9)
- **Agent memory** persisted per business (recent decisions, preferences, past approvals)
- **RAG** introduced once there is enough business-specific unstructured content (uploaded docs, past campaigns) to justify retrieval
- **Model routing** across multiple providers with fallback and free/paid tiering for cost control
- **Token/context optimization** and prompt management become dedicated internal concerns as agent count grows

Explicitly: none of the routing, memory, RAG, or multi-provider complexity is required to ship MVP. Building it prematurely would violate the “start small, architect for scale” principle (Section 5).

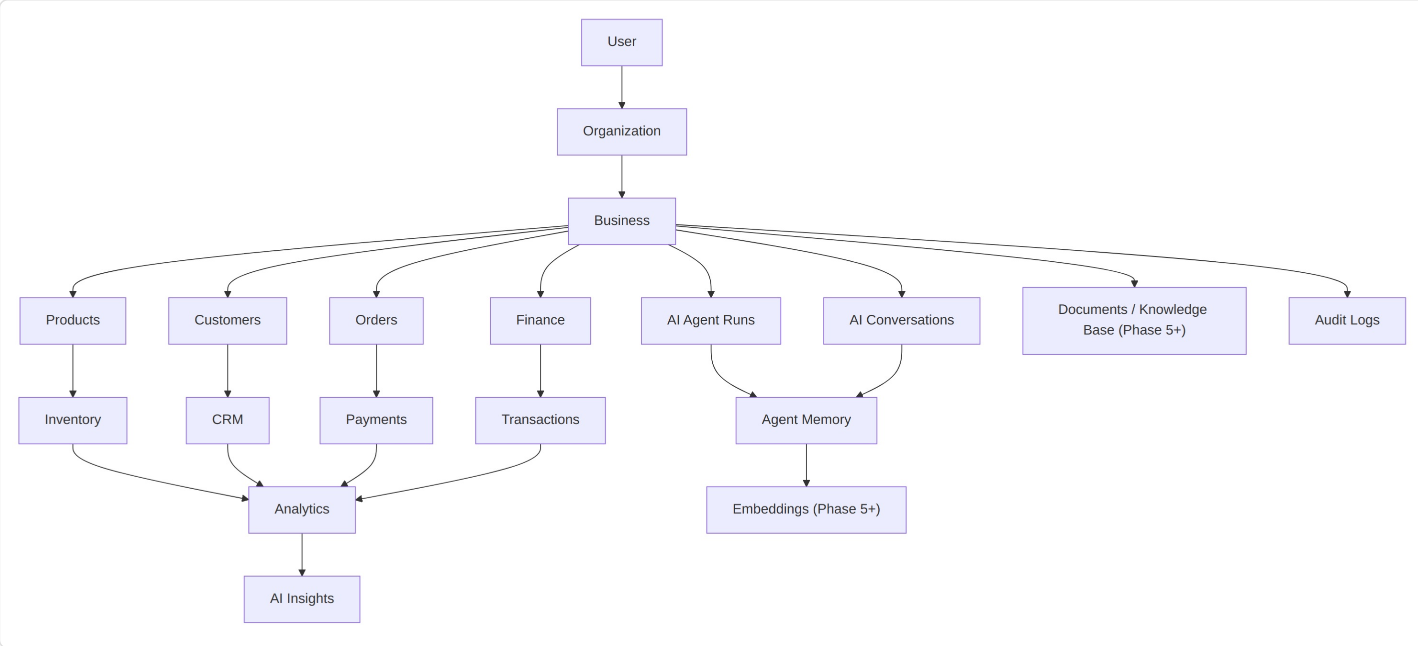
31. AI Agent Map



31.1 Agent-to-Capability Mapping

Agent	Tools	Data Access	Permissions	Actions	Human Approval
Setup Agent	Idea parser, profile writer	BusinessProfile (write draft)	Draft-only	Populate profile fields	Required to activate
Research Agent	Category/competitor lookup	BusinessProfile (read)	Read-only + draft notes	Attach research summary	Not required (informational)
Product Content Agent	Copy generator, image prompt	Product (read/write draft)	Draft-only	Populate description fields	Required to publish
Marketing Agent	Campaign generator	Product, BusinessProfile (read)	Draft-only	Create MarketingCampaign draft	Required to publish/schedule
Sales Agent	Follow-up drafter	Customer, Order (read)	Draft-only	Draft outreach message	Required to send
Inventory Agent	Stock analyzer	Inventory (read), alerts (write)	Read + alert-write	Create low-stock Notification	Required to trigger reorder
Finance Agent	Invoice drafter, anomaly detector	Invoice, Transaction (read/draft)	Draft-only	Draft Invoice	Required to send/finalize
Analytics Agent	Metric summarizer	Cross-module read	Read-only	Generate insight text	Not required (informational)

32. Data Architecture



What belongs where:

- **Operational data** (Products, Customers, Orders, Finance) is the system of record — always tenant-scoped to `Business`.
- **Analytics** is derived, not authoritative — it's computed from operational data, never a separate source of truth.
- **AI Insights** are generated text/recommendations layered on top of Analytics — stored for history but always regenerable.
- **AI Agent Runs / Conversations / Memory** are AI-specific operational data, tenant-scoped the same way, and feed the Audit Log.
- **Embeddings / Knowledge Base** are Phase 5+ additions once RAG is justified (Section 30.2) — not part of MVP data architecture.
- **Audit Logs** span the whole system and are append-only.

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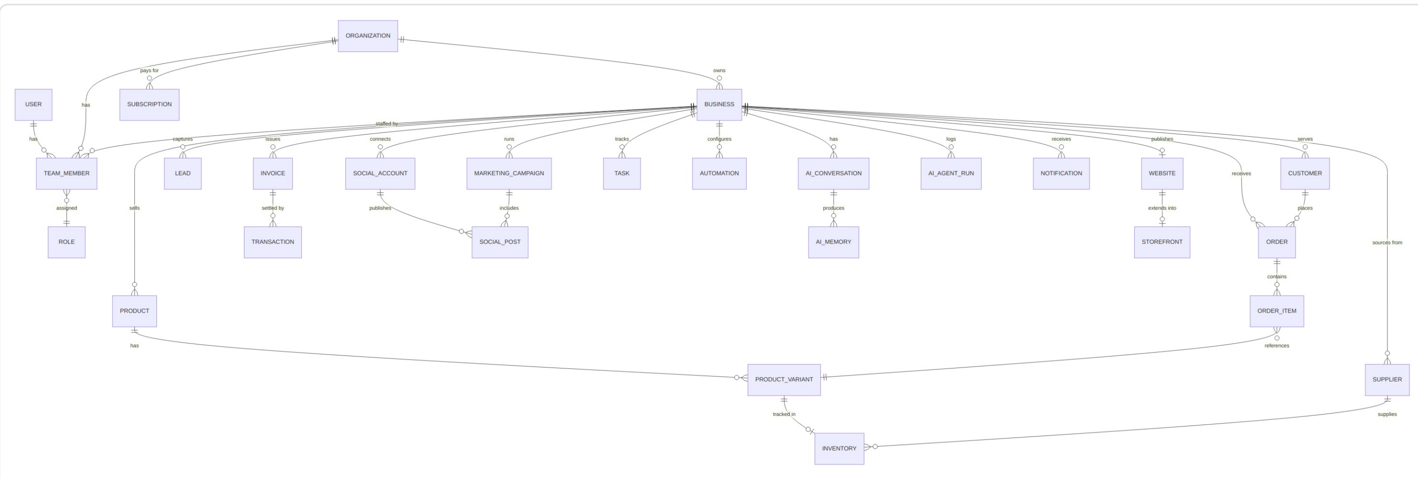
33. Database ER Diagram

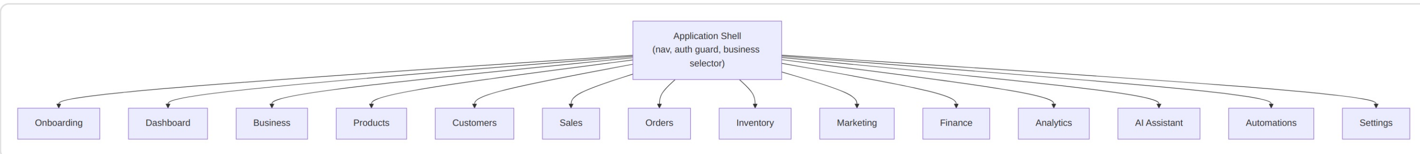


33.1 Entity Notes

Entity	Purpose	Key Fields	Tenant Isolation
USER	Login identity	email, auth_id	Global (not tenant-scoped)

ORGANIZATION	Top-level billing/tenant unit	name, plan	Root of isolation tree
BUSINESS	The operated business	name, category, org_id	Primary isolation key for all downstream tables
TEAM_MEMBER	Links User↔Business with a Role	user_id, business_id, role_id	Scoped to business_id
ROLE	Permission set	name, permissions	Global reference table
PRODUCT / PRODUCT_VARIANT	Sellable items	name, price, business_id	Scoped to business_id
INVENTORY	Stock levels	variant_id, quantity	Scoped via variant → business
SUPPLIER	Inventory source	name, business_id	Scoped to business_id
CUSTOMER / LEAD	People the business serves	name, contact, business_id	Scoped to business_id
ORDER / ORDER_ITEM	Sales transactions	customer_id, total, status	Scoped to business_id
INVOICE / TRANSACTION	Financial records	amount, status	Scoped to business_id
MARKETING_CAMPAIGN	Campaign content/schedule	name, content, business_id	Scoped to business_id
SOCIAL_ACCOUNT / SOCIAL_POST	Connected channels/content	platform, business_id	Scoped to business_id
TASK	Owner/AI action items	title, status, business_id	Scoped to business_id
AUTOMATION	Trigger→action rules	trigger, action, business_id	Scoped to business_id
AI_CONVERSATION / AI_AGENT_RUN	Agent interaction history	agent_type, input, output	Scoped to business_id, feeds Audit
AI_MEMORY	Persisted agent context (Phase 5+)	summary, embedding	Scoped to business_id
NOTIFICATION	Alerts/reminders	message, read_state	Scoped to business_id
SUBSCRIPTION	Billing plan state	plan, status	Scoped to organization_id
WEBSITE / STOREFRONT	Public presence (Phase 7+)	template, domain	Scoped to business_id

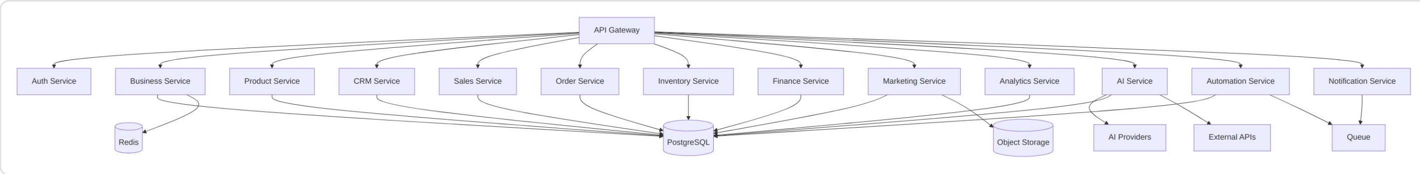
34. Frontend Architecture



Routing & component notes:

- The Application Shell owns authentication guarding, the active-business context, and top-level navigation; every module below it is a routed sub-tree.
- Each module (Products, Customers, Orders, etc.) is a self-contained feature folder: its own routes, list/detail components, and API hooks — enabling modules to be shipped or hidden independently per Principle 5 (modular architecture).
- The **AI Assistant** is not a separate “page” only — it is also embedded contextually inside other modules (e.g., a “Generate description” button inside Products calls the same underlying agent as the standalone Assistant screen).
- Modules not yet built (Inventory, Marketing, Finance, Automations pre-Phase 6/8) are simply absent from the sidebar until their phase ships — no placeholder/“coming soon” screens in MVP.

35. Backend Architecture



MVP recommendation — modular monolith:

For MVP, every "service" box above (Business, Product, CRM, Sales, Order, AI) should be a **module inside a single deployable backend**, not a separately deployed microservice. Reasons:

- A single small team cannot operate 8+ independently deployed services without disproportionate DevOps overhead.
- Tenant-scoped data access patterns (everything keyed by `business_id`) are simpler to enforce consistently inside one codebase.
- Module boundaries are enforced at the **code/package level** (clear internal interfaces), so splitting into real microservices later — if and when a specific module needs independent scaling (most likely the **AI Service**, due to different latency/cost/scaling characteristics) — is a refactor, not a rewrite.

Explicitly avoid: standing up separate microservices for Business/Product/CRM/Sales/Order at MVP. This is over-engineering relative to actual load and team size. The **AI Service** is the one component worth keeping architecturally separable early, since LLM calls have fundamentally different latency, cost, and scaling profiles than CRUD operations.

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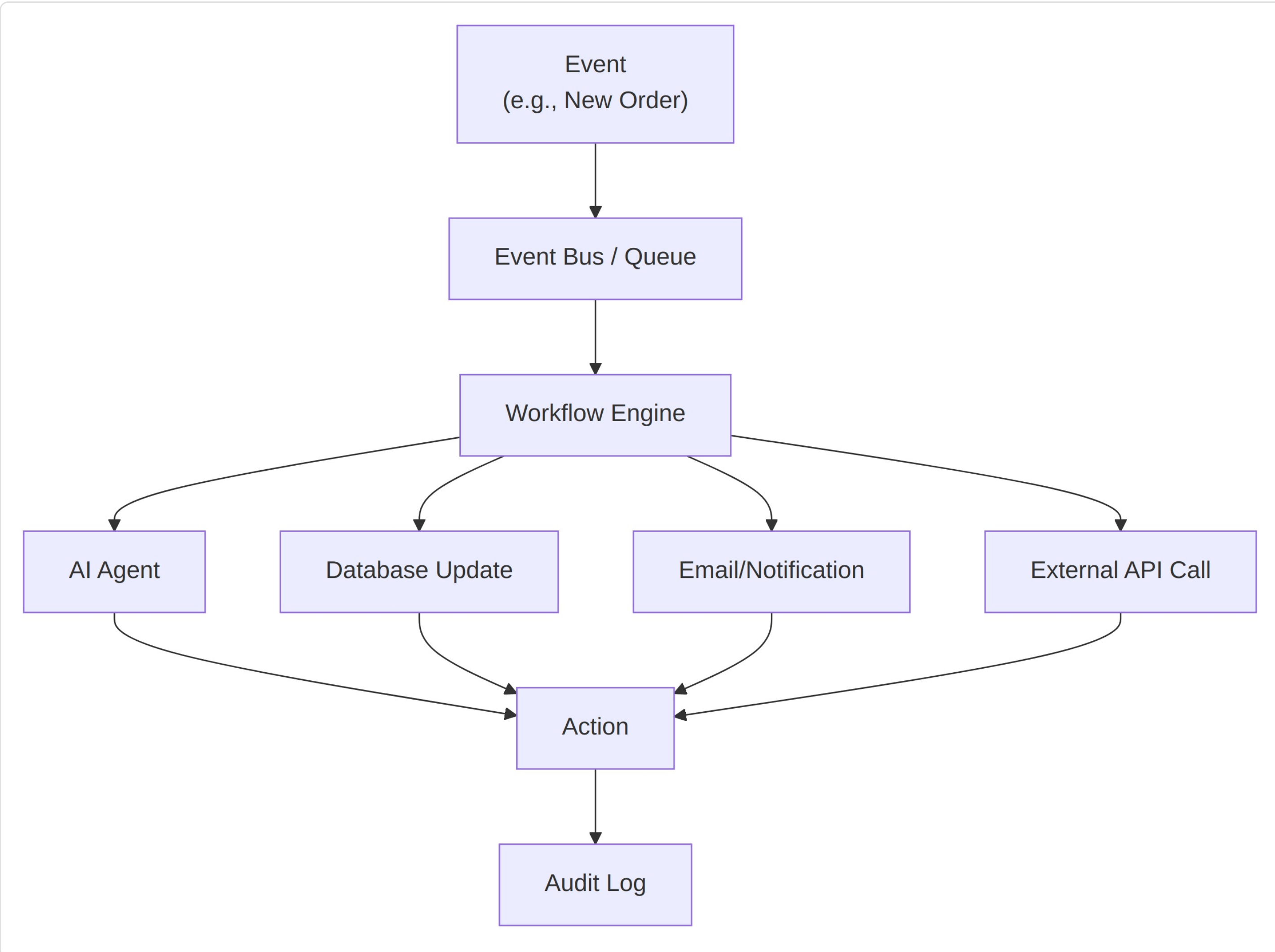
36. Automation Architecture



Worked examples:

- **New Order:** Order created → Inventory updated → Customer record updated → Revenue recorded → Owner notified → Analytics Agent generates a business insight.
- **Low Inventory:** Threshold detected → Owner notified → Inventory Agent suggests a reorder quantity (approval-gated before any purchase action).
- **New Product:** Description generated (Product Content Agent) → Social content drafted (Phase 5) → Campaign prepared (Phase 5) → Owner approval requested before anything publishes.

Per Section 9.4/18, only the internal, reversible steps (inventory update, record update, internal notification, draft generation) run automatically; anything customer-facing or irreversible pauses for approval.

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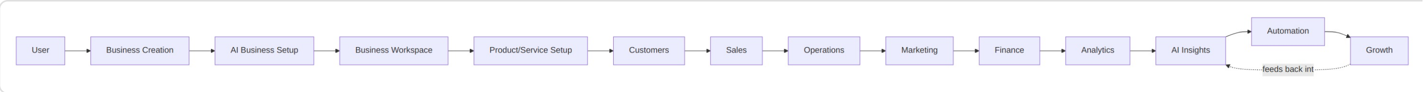
37. Complete Product Flow (Master Diagram)



This is the single diagram that should anchor every other conversation about the product — every module and agent in this document exists to serve one segment of this loop.

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38. What We Are Actually Building

Plainly, for an engineer starting tomorrow:

We are building a **multi-tenant SaaS web application**.

A user creates an account. The user creates a business workspace. The user describes their business in plain text. BusinessOS stores this as a business profile record. An AI agent analyzes that description and produces structured recommendations (positioning, target customer, initial plan). The user reviews and edits those recommendations. The user creates product or service records. The platform stores all of this as normal relational data, scoped to that business. The dashboard queries and aggregates that data. AI agents can read business data they've been granted access to, and can call a defined set of tools/actions — never arbitrary ones. Automations are simple trigger→action rules that react to business events (new order, low stock, new product). The business owner can approve or reject any AI-proposed action that is customer-facing, financial, or irreversible. Every AI action, whether autonomous or approved, is logged. Later, the platform provides a public storefront so the business's own customers can browse and order directly, with those orders flowing back into the same workspace.

Capability	MVP	Phase 2-3	Future
Account/business creation	✅	—	—
AI-drafted business setup	✅ (basic)	Deeper research/positioning	Continuous re-analysis
Product/service records + AI content	✅	Image generation, variants	Bulk import, catalogs
CRM (customer records)	✅ (basic)	Lead pipeline, notes/history	Segmentation, lifecycle automation
Orders	✅ (manual entry)	Status pipeline	Storefront-originated orders
Dashboard	✅ (basic metrics)	AI insight narratives	Predictive recommendations
Marketing content	✅ (copy only)	Scheduling, social publishing	Multi-channel campaign orchestration
Inventory	—	Phase 6	Supplier integration, forecasting
Finance	—	Phase 6	Full bookkeeping, tax support
Automation engine	—	Basic triggers (Phase 4–5)	Full rule builder (Phase 8)
Storefront	—	—	Phase 7

Marketplace	—	—	Phase 10

39. Expected Final Product

A new entrepreneur opens BusinessOS and sees: **“Let’s build your business.”**

They describe their idea in a sentence. BusinessOS creates their workspace and, within moments, shows a dashboard summarizing Business Health, Revenue, Orders, Customers, Products, Tasks, Marketing performance, AI Recommendations, and (later) Automations — all populated from real data, not placeholders.

The AI assistant understands the business’s actual context, so instead of generic prompts, the owner can ask things like:

- “How is my business doing?”
- “What should I focus on today?”
- “Which products are performing best?”
- “Why did sales drop this week?”
- “Create a marketing campaign for my best-selling product.”
- “Show me customers who haven’t purchased recently.”
- “What inventory do I need to reorder?”

Every answer is grounded in that business’s actual stored data — orders, products, customers, campaigns — not a generic LLM response. This is the experience the entire architecture in this document exists to make possible.

40. Page-by-Page Product Blueprint

The following blueprint covers every major screen. MVP-critical screens are specified in full; later-phase screens are specified to the same standard but marked with their phase.

40.1 Dashboard (MVP)

Aspect	Detail
Purpose	Daily home view answering “what’s happening / what’s next”
Access	Any authenticated Team Member of the active Business
Main components	Business Health cards, AI Recommendation panel, Task list, Recent Orders, Recent Activity
Data displayed	Revenue trend, order count, customer count, top AI recommendation, open tasks
User actions	Approve/dismiss AI recommendation, complete task, switch business (if multiple)
AI actions	Advisor Agent generates ranked recommendations; Analytics Agent (Phase 3+) generates insight text
API calls	GET /dashboard/summary, GET /tasks, GET /ai/recommendations
Loading state	Skeleton cards for each section
Empty state	“Your business is just getting started — here’s what to do first,” with a direct link into setup
Error state	Inline retry banner per failed section (partial failure shouldn’t blank the whole page)
Success state	Populated cards with live data
Mobile behavior	Cards stack vertically; recommendation panel remains top-most
Desktop behavior	Multi-column grid layout

Permissions	Read-only for view; task/recommendation actions require Owner or permitted Team Member role
Analytics events	dashboard_viewed, recommendation_approved, recommendation_dismissed, task_completed

40.2 Products — List & Detail (MVP)

Aspect	Detail
Purpose	Create and manage sellable products/services
Access	Owner, Team Members with product-edit permission
Main components	Product list (table/cards), create/edit form, AI content panel
Data displayed	Name, price, category, status (draft/published), stock (once Inventory ships)
User actions	Create, edit, delete, publish/unpublish product
AI actions	Product Content Agent drafts description, title variants, bullet features
API calls	GET/POST/PATCH/DELETE /products, POST /ai/product-content
Loading state	Skeleton rows/cards
Empty state	“No products yet — create your first one,” with CTA into the create form
Error state	Field-level validation errors on form; list-level retry banner
Success state	Toast confirmation + updated list
Mobile behavior	Card list, full-screen create/edit form
Desktop behavior	Table list, side-panel or modal create/edit form
Permissions	View: all Team Members; Edit/Publish: Owner + permitted roles
Analytics events	product_created, product_published, ai_content_generated, ai_content_accepted

40.3 Customers — List & Detail (MVP)

Aspect	Detail
Purpose	Basic CRM — track who the business’s customers are
Access	Owner, Team Members with CRM permission
Main components	Customer list, customer detail (contact info, order history, notes)
Data displayed	Name, contact, total orders, last order date
User actions	Add/edit customer, add note, view order history
AI actions	CRM Agent flags at-risk/inactive customers (Phase 4+)
API calls	GET/POST/PATCH /customers, GET /customers/{id}/orders
Loading state	Skeleton list
Empty state	“No customers yet — they’ll appear here as you record sales”
Error state	Inline retry
Success state	Updated list/detail view
Mobile behavior	Stacked list, drill-in detail view
Desktop behavior	List + detail split view

Permissions	View/edit gated by role once Team Management ships (Phase 4)
Analytics events	customer_created, customer_note_added

40.4 Orders (MVP — manual entry)

Aspect	Detail
Purpose	Record and track sales transactions
Access	Owner, Team Members with sales permission
Main components	Order list, order detail, manual order-entry form
Data displayed	Order items, total, customer, status, date
User actions	Create order, update status, view detail
AI actions	None in MVP; Analytics Agent references order data (Phase 3+)
API calls	GET/POST/PATCH /orders
Loading/Empty/Error/Success	Standard patterns as above
Mobile/Desktop	List + detail, same pattern as Customers
Permissions	Create/edit gated by role
Analytics events	order_created, order_status_updated

40.5 AI Assistant (MVP — basic)

Aspect	Detail
Purpose	Direct conversational access to the Business Advisor Agent
Access	Owner and permitted Team Members
Main components	Chat interface, suggested prompts, action confirmation cards
Data displayed	Conversation history (session-scoped in MVP)
User actions	Ask a question, approve/reject a suggested action
AI actions	Advisor Agent responds using live business data (Section 39)
API calls	POST /ai/assistant/message
Loading state	Typing indicator
Empty state	Suggested starter prompts (“Ask me how your business is doing”)
Error state	Inline “couldn’t complete that — try again” message
Success state	Response rendered with any approval-required action shown as a distinct card
Mobile/Desktop	Full-screen chat on mobile; chat + business context panel on desktop
Permissions	Approval-gated actions restricted to Owner by default
Analytics events	assistant_message_sent, assistant_action_approved, assistant_action_rejected

(Later-phase screens — Inventory, Marketing/Campaigns, Finance, Analytics detail, Automations, Notifications, Settings — follow the identical blueprint structure above and should be specified in full at the start of their respective phase, not before, to avoid designing against data that doesn’t exist yet.)

---

41. Visual UI Wireframe Descriptions

41.1 Dashboard

```
------------------------------------------------
Top Navigation
Business Selector | Search | Notifications | User
------------------------------------------------
Sidebar                 Main Content
Dashboard                Good morning, [Business Name]
Business
Products                 Business Health
Customers                [Revenue] [Orders] [Customers] [Profit]
Orders
Inventory                ------------------------------------------
Marketing                 AI RECOMMENDATION
Finance                   "Your best-selling product is running low.
Analytics                  Would you like me to prepare a reorder?"
AI Assistant               [Review]  [Automate]
Automations               ------------------------------------------
Settings                   Today's Tasks
                           - ...
                           ------------------------------------------
                           Recent Orders
                           - ...
------------------------------------------------
```

41.2 Products List

```
------------------------------------------------
Top Navigation
------------------------------------------------
Sidebar                 Main Content
                          Products                      [+ New Product]
                          ------------------------------------------
                          [Search products...]     [Filter: Status ▾]
                          ------------------------------------------
                          Name          Price    Status     Stock
                          Cotton Kurta  ₹899     Published   -
                          Blue Saree    ₹1499    Draft       -
                          ------------------------------------------
```

41.3 Product Create/Edit (with AI panel)

```
------------------------------------------------
New Product
------------------------------------------------
Name: [________________]        AI CONTENT PANEL
Price: [______]                  "Generate description"
Category: [____________]         [Generate]
                                  -------------------------
Description:                     Draft:
[__________________________]     "Soft cotton kurta with a
[__________________________]      relaxed fit, perfect for..."
                                  [Use this]  [Regenerate]
[Save Draft]  [Publish]
------------------------------------------------
```

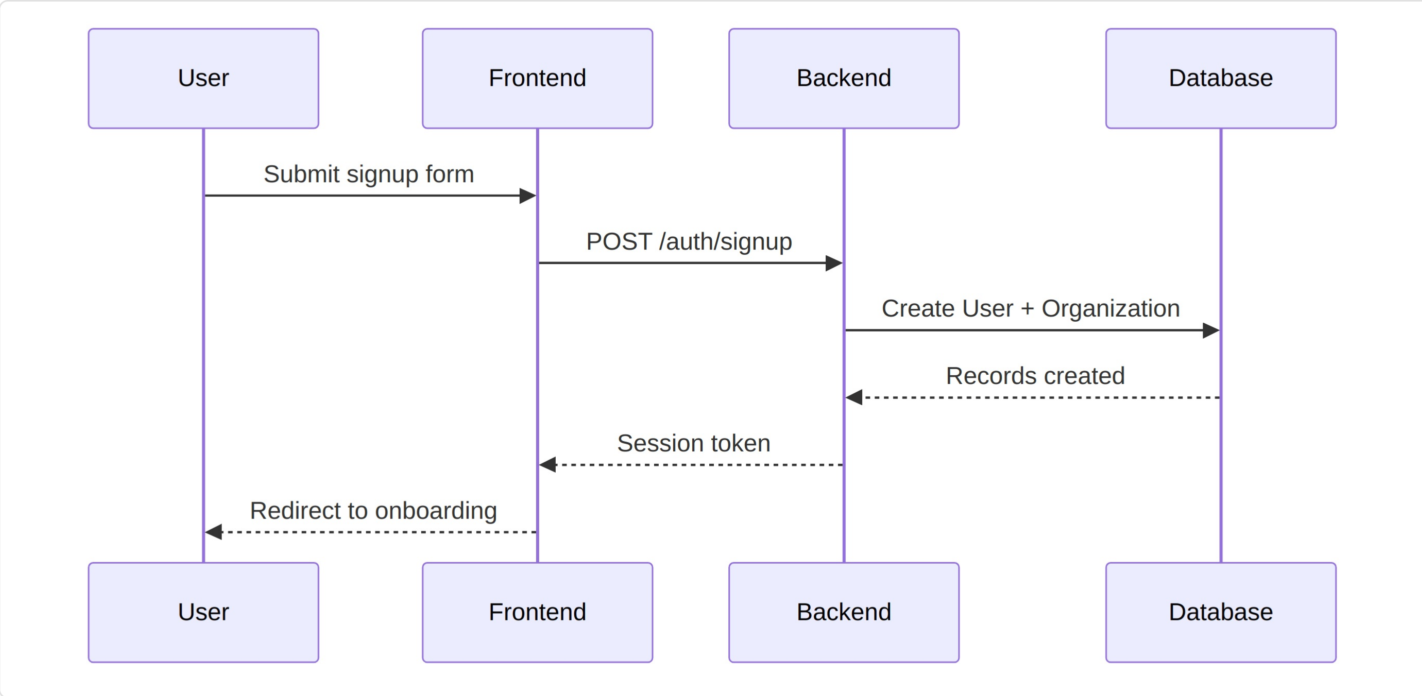
41.4 AI Assistant

```
------------------------------------------------
AI Assistant
------------------------------------------------
```

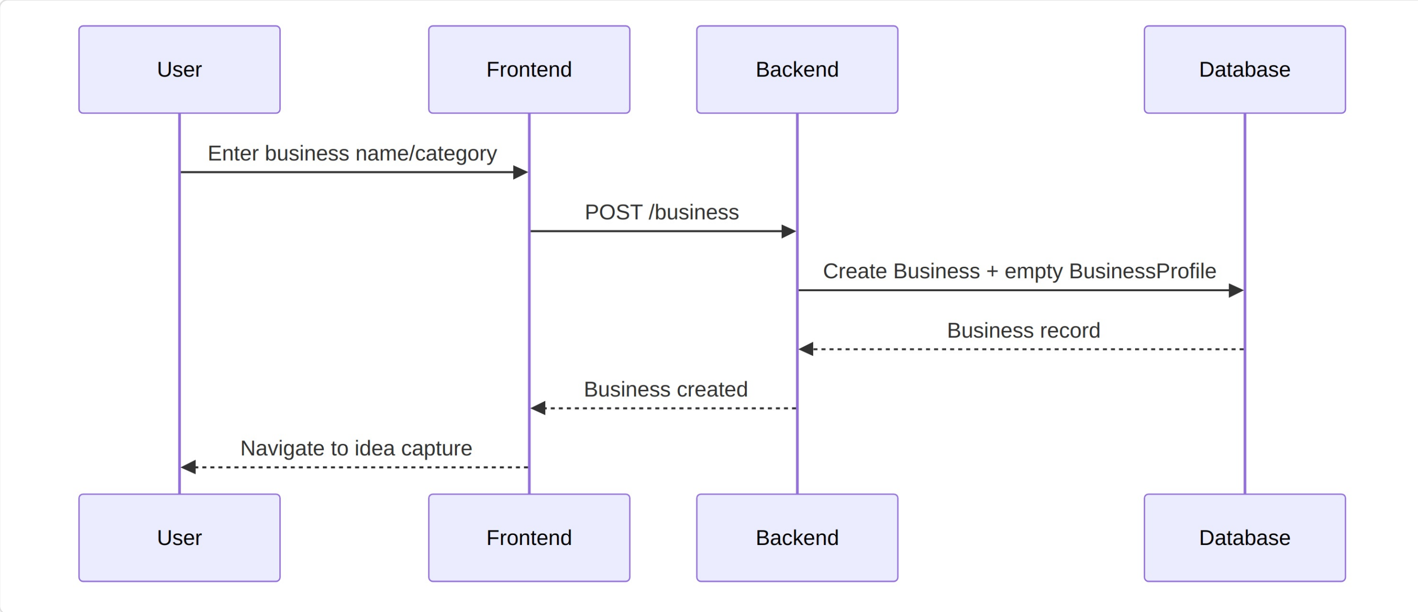
```
"How is my business doing?"

BusinessOS: Revenue is up 12% this week, driven mainly
by your Cotton Kurta line. 3 customers haven't ordered
in 30+ days — want me to draft a follow-up?

[Draft follow-up]    [Not now]
------------------------------------------------
[Type a message...]                        [Send]
------------------------------------------------
```

42. System Sequence Diagrams

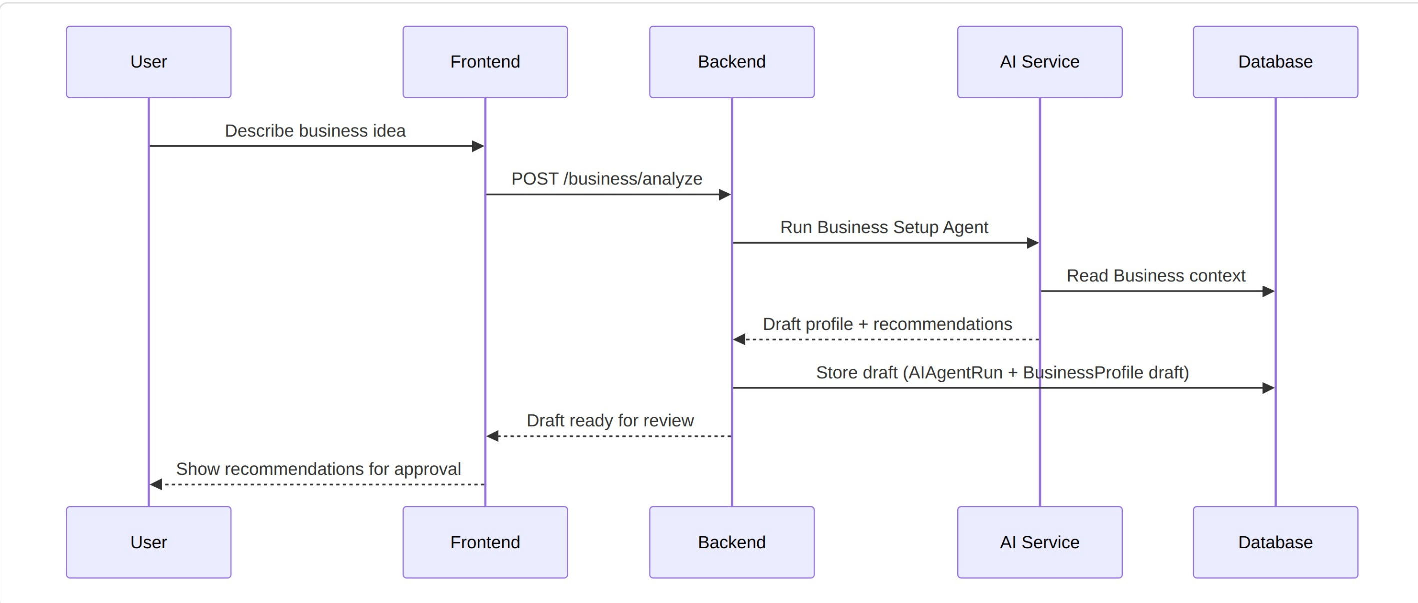
42.1 User Signup



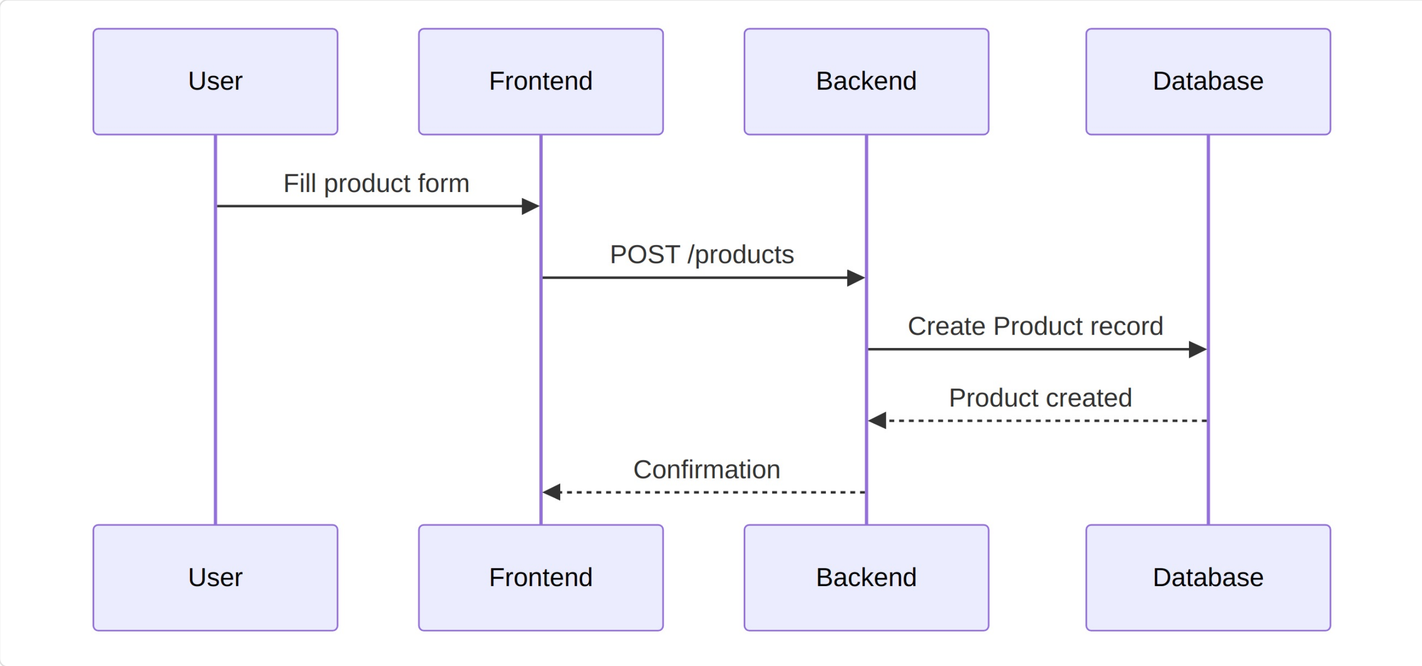
42.2 Business Creation



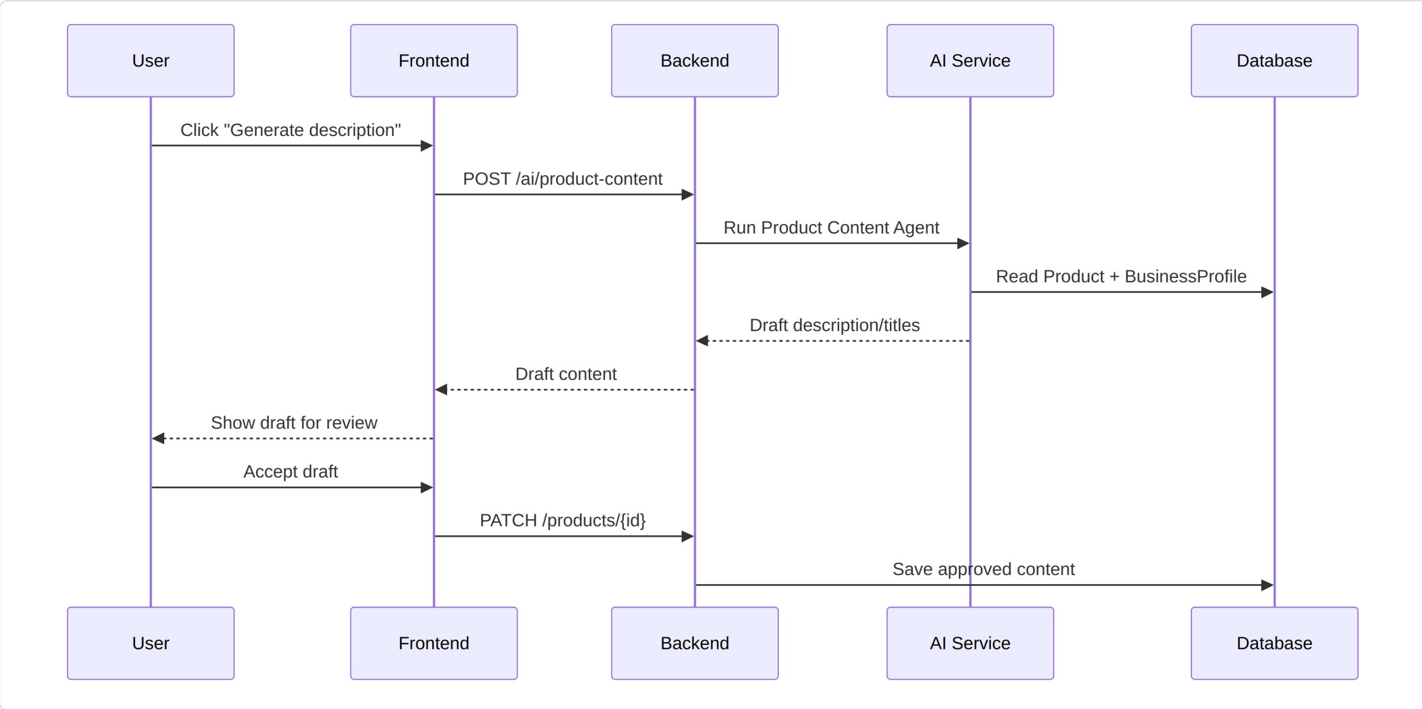
42.3 AI Business Analysis



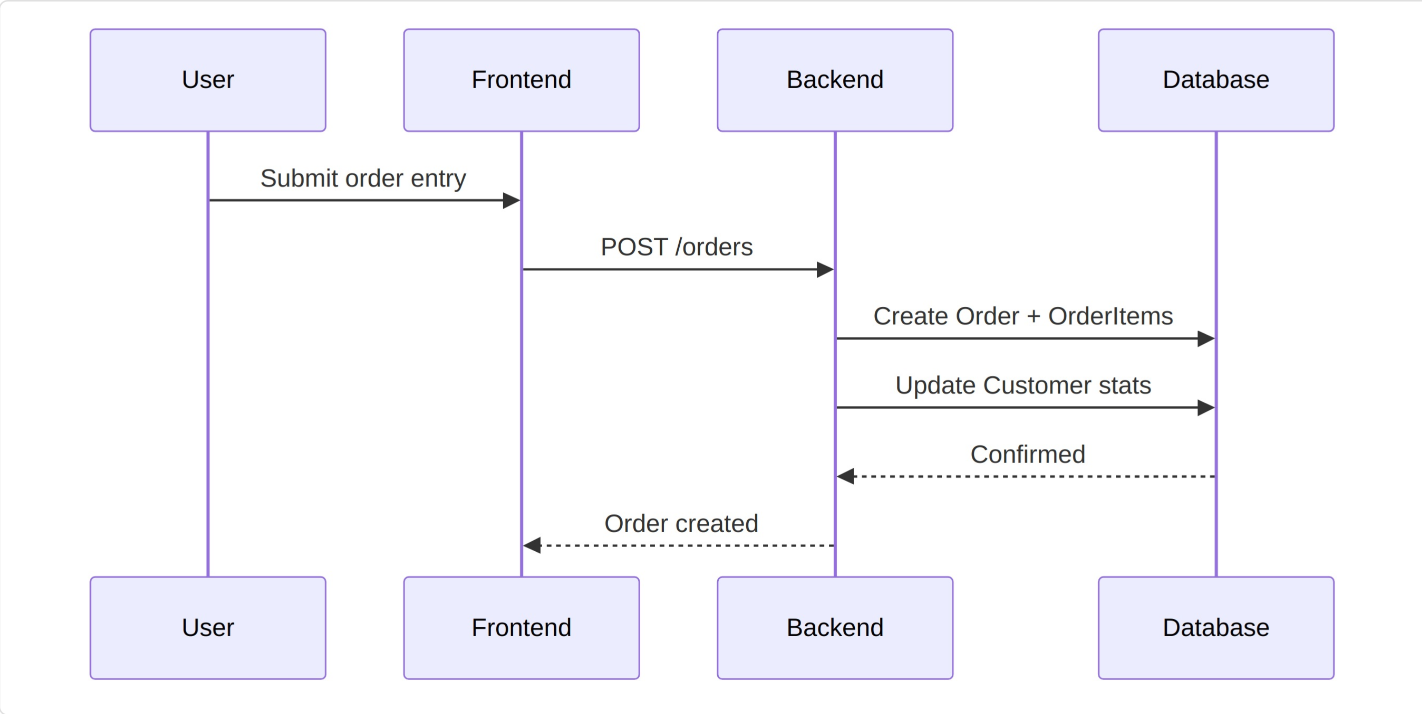
42.4 Product Creation



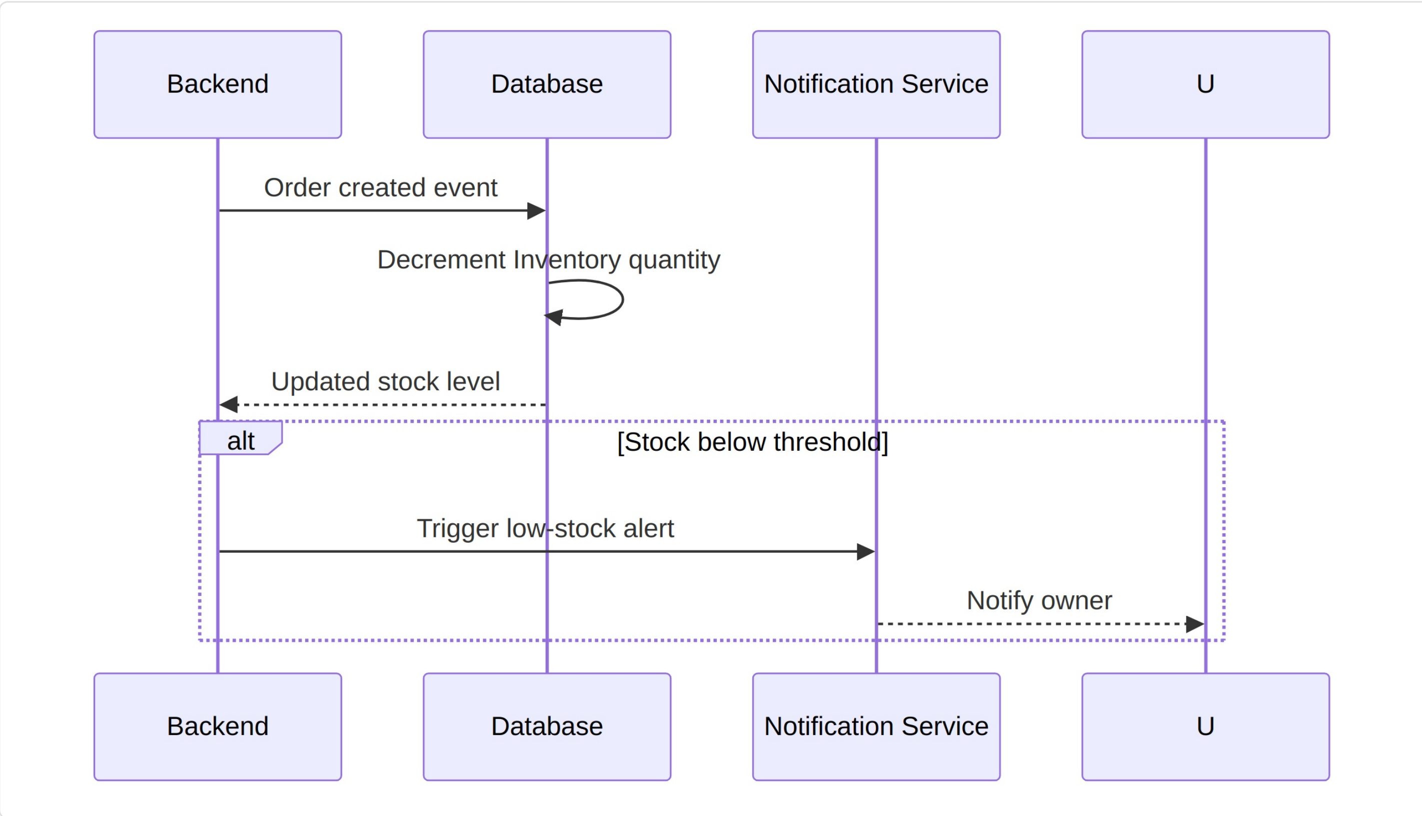
42.5 AI Content Generation



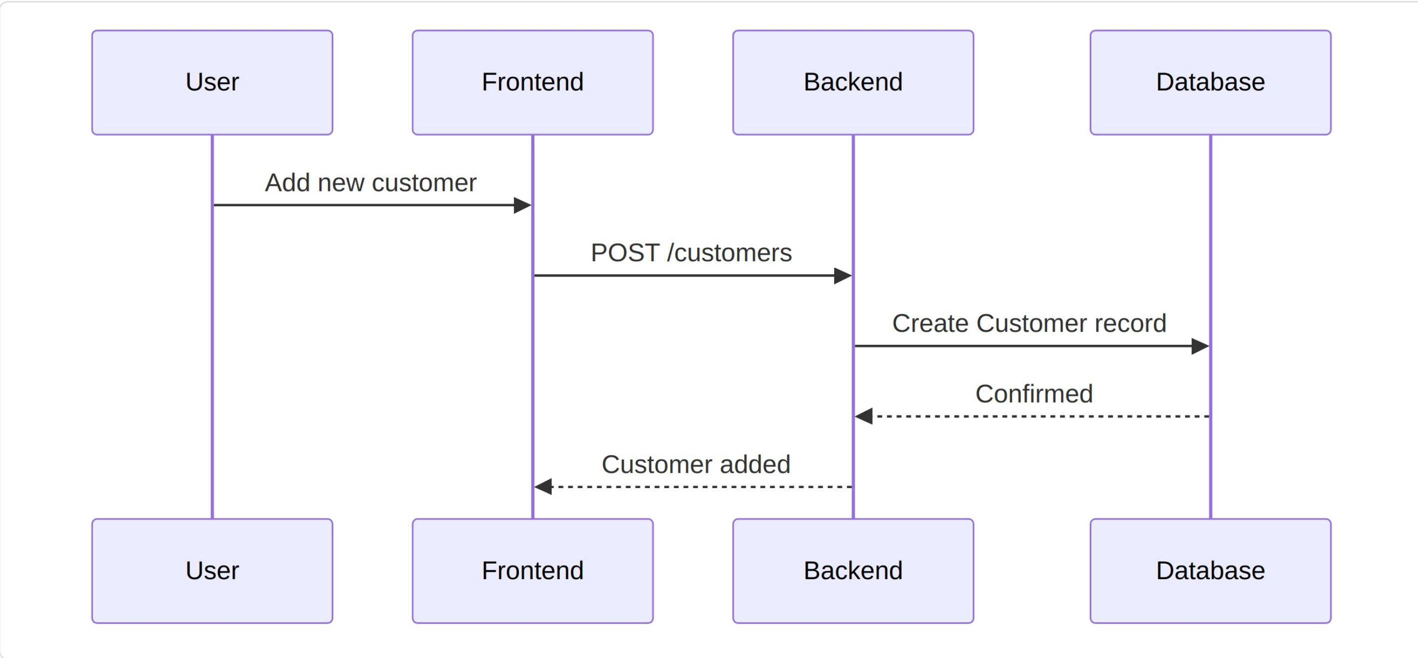
42.6 New Order



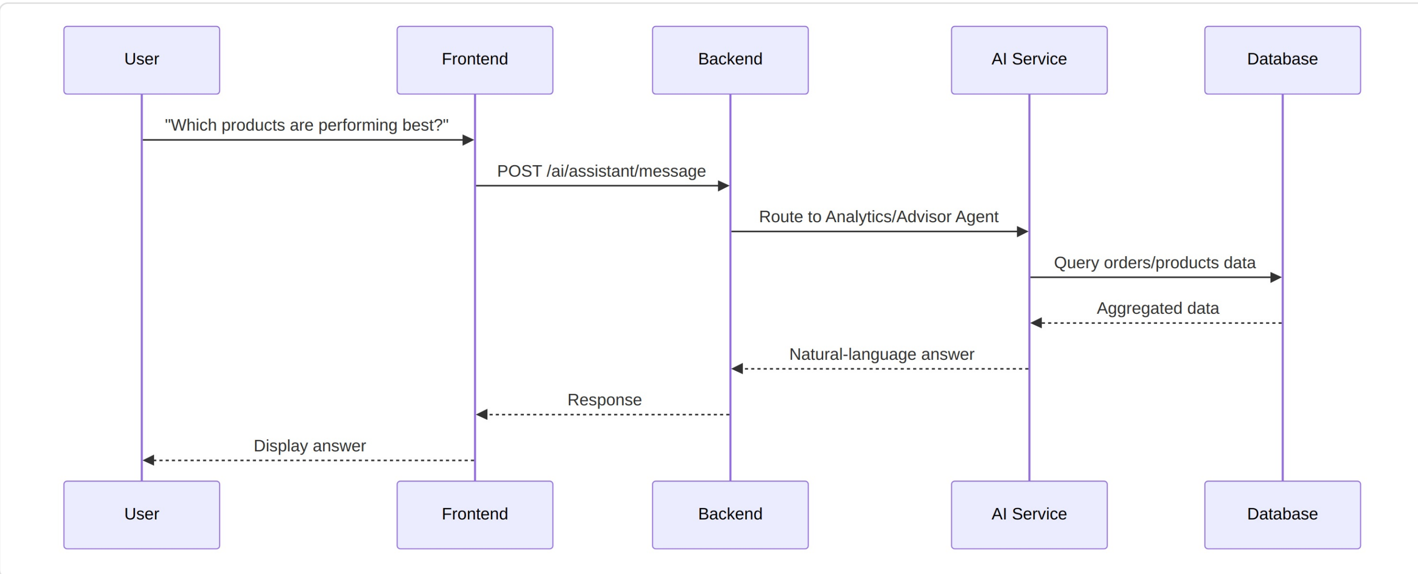
42.7 Inventory Update (Phase 6)



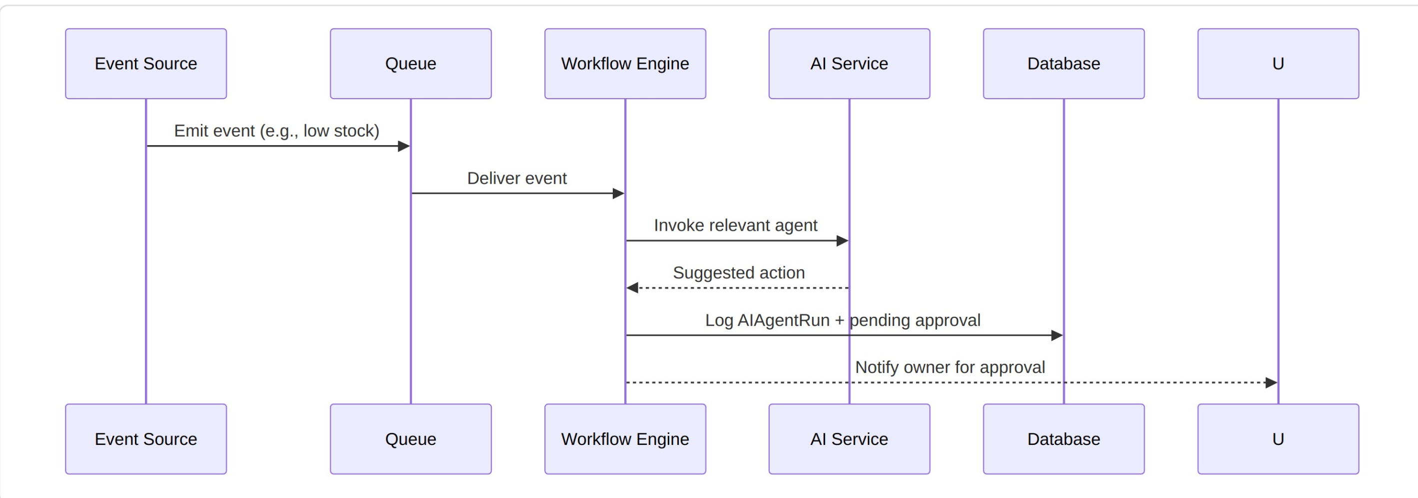
42.8 Customer Creation



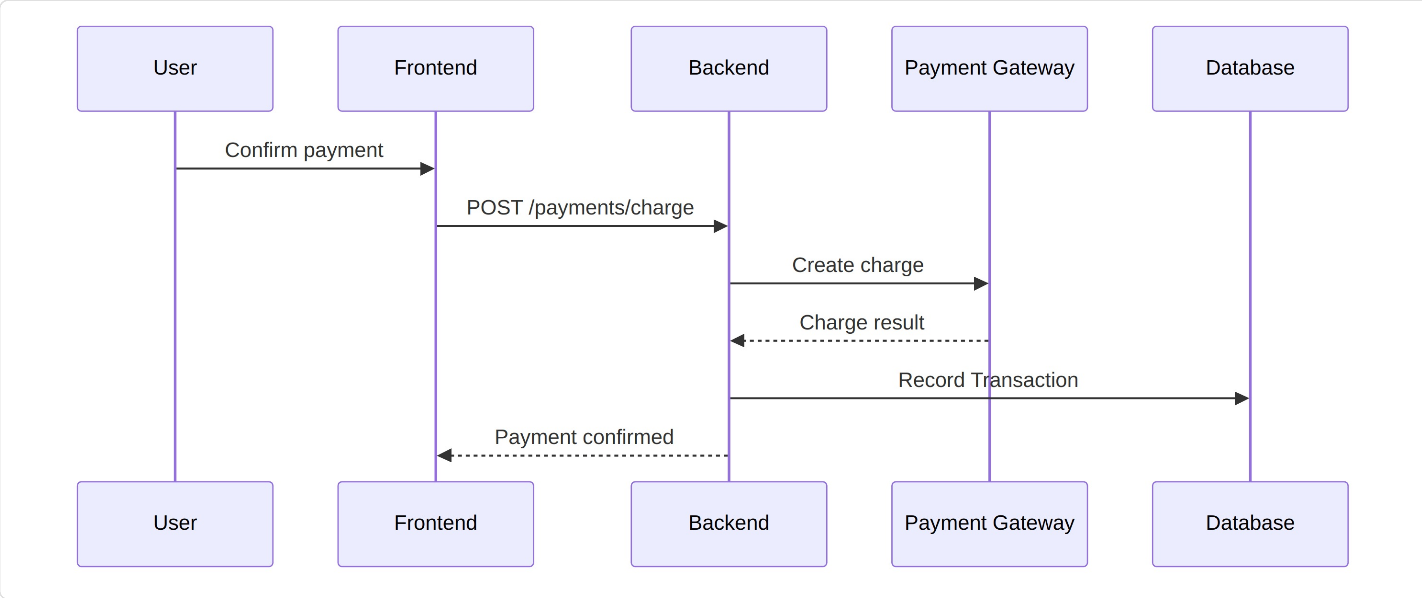
42.9 AI Business Query (Assistant)



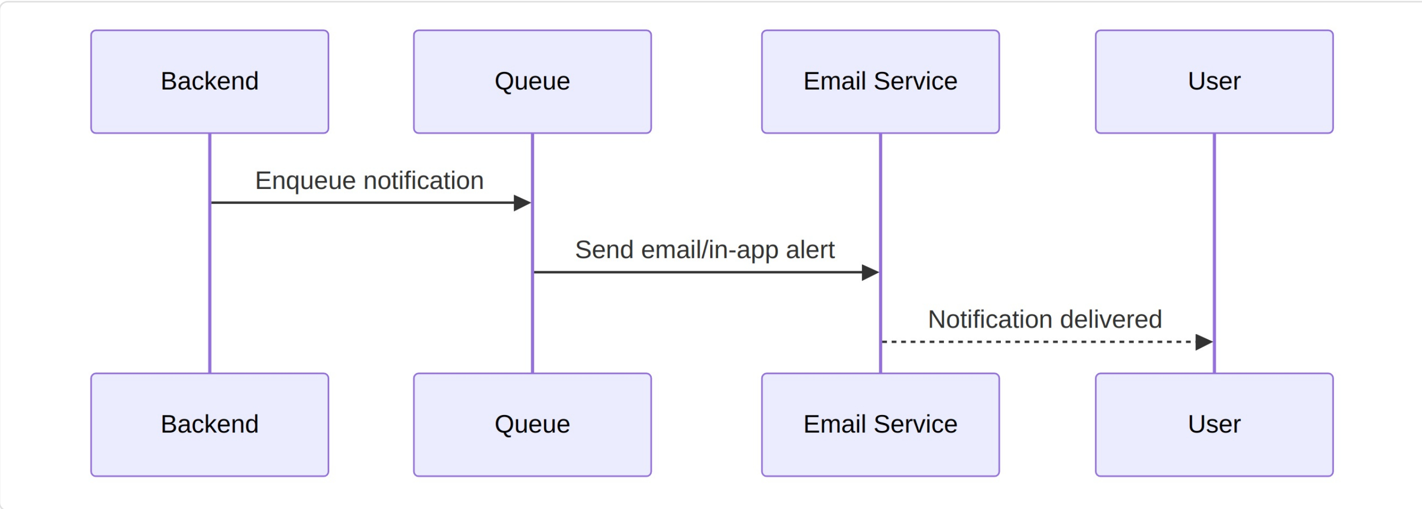
42.10 Automation Execution (Phase 8)



42.11 Payment (Phase 6+)

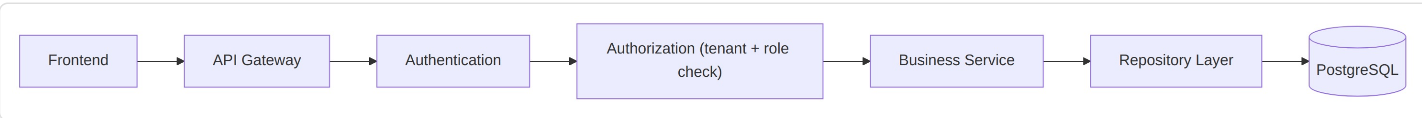


42.12 Notification

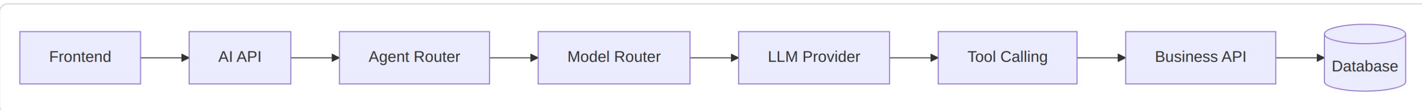


43. API Flow

Standard business data flow:



AI-involved flow:



Step explanations:

1. **Authentication** confirms who the caller is.

2. **Authorization** confirms they may act on this specific Business (tenant check) and have the required role/permission.
3. **Business Service** contains the actual logic (validation, side effects).
4. **Repository Layer** is the only code allowed to issue SQL/ORM queries — services never query the database directly, keeping tenant-scoping enforcement centralized.
5. For AI flows, the **Agent Router** decides which specialized agent handles the request; the **Model Router** picks the LLM provider/tier (MVP: a single fixed provider, no real routing logic yet, per Section 30.1); **Tool Calling** is how the agent reaches back into normal Business APIs — an agent never touches the database directly, only through the same service/repository layers as the frontend.

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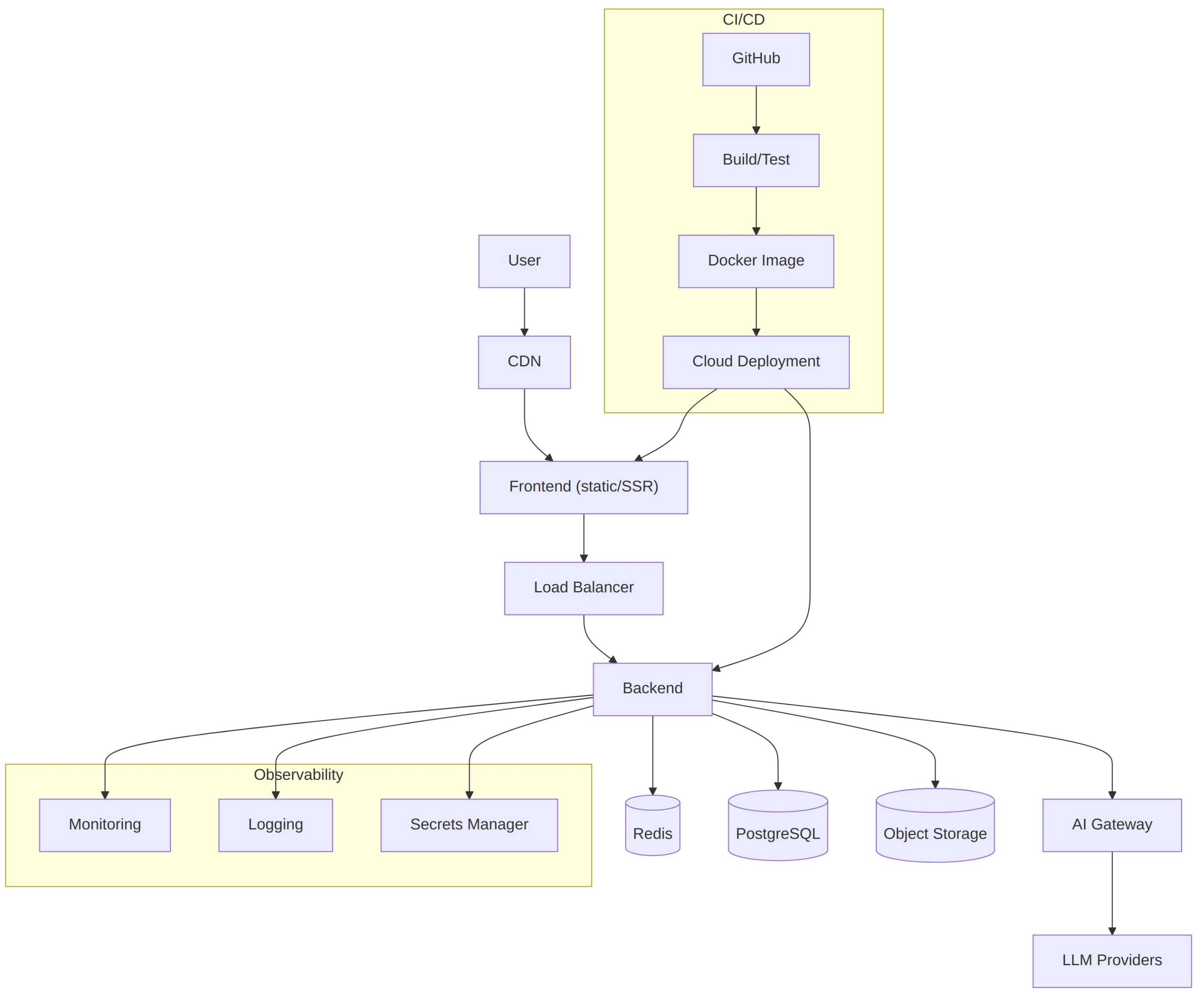
44. Deployment Architecture



MVP deployment should stay simple: one frontend deployment, one backend deployment (modular monolith per Section 35), managed Postgres/Redis, a single CI/CD pipeline (GitHub Actions or equivalent), and baseline monitoring/logging — not a multi-region, multi-cluster setup.

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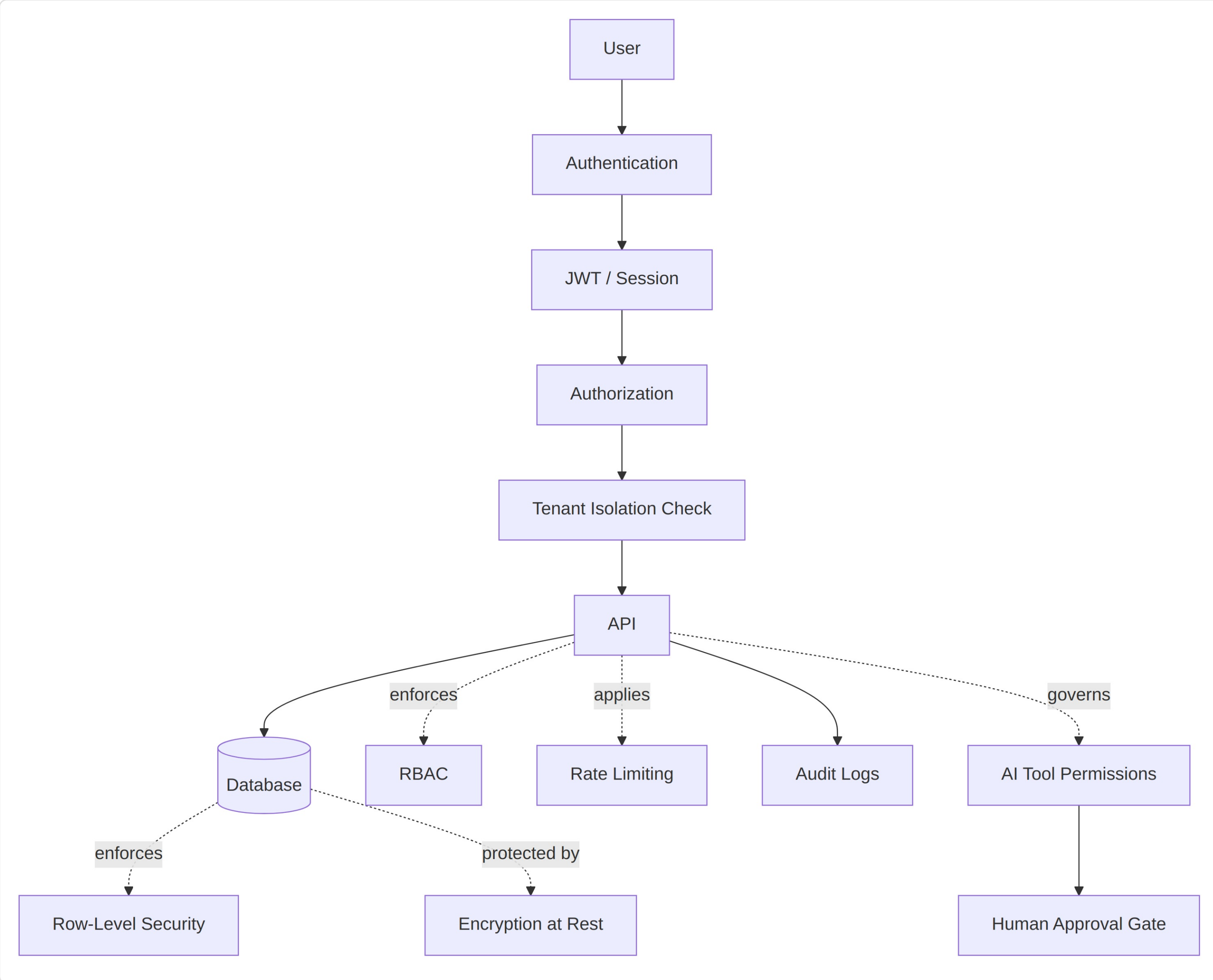
45. Security Architecture



Every request passes: Authentication → Session validation → Authorization → Tenant isolation check, before it ever reaches business logic or the database. AI tool calls pass through the same Authorization and Tenant Isolation checks as any other API caller — an agent has no special bypass — and additionally pass through the Approval Gate defined in Section 9.4 for any non-autonomous-tier action.

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46. Visual Requirements Note

All diagrams in this document are provided as Mermaid diagrams, which render directly wherever Mermaid is supported and remain readable as plain text otherwise. No decorative or non-explanatory imagery has been included — every visual above communicates architecture, data flow, user flow, or UI structure directly relevant to building BusinessOS AI.

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47. Final Outcome — What We Will Have

After the full roadmap is completed, BusinessOS AI will provide:

1. Business creation
2. Business onboarding
3. AI business advisor
4. Business dashboard

5. Product/service management
6. Customer management
7. Sales management
8. Order management
9. Inventory
10. Finance
11. Marketing
12. AI content generation
13. AI agents (full roster)
14. Automation
15. Analytics
16. Notifications
17. Team management
18. Integrations
19. Website/storefront
20. Future marketplace

Core MVP = items 1–8, 12 (basic), and 13 (two agents only), 15 (basic), 16 (basic) — everything scoped explicitly in Section 19.

Future Platform = items 9–11, 14, and the full depth of 13/15/17–20, delivered across Phases 2–10 (Section 20).

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48. CTO Build Recommendation — Milestones

MILESTONE 0 — Product Foundation *Outcome:* Finalized specification (this document), confirmed technical architecture, completed SMART-BUSSINESS repo audit. Features: none (documentation only). Screens: none. Database: none. APIs: none. AI: none. Tests: none. Definition of Done: this document reviewed and signed off by all stakeholders. Expected User Outcome: n/a. Expected Business Outcome: shared clarity before spend begins.

MILESTONE 1 — Identity + Business *Outcome:* A user can sign up, create an organization and a business, and land in an empty workspace. Features: signup/login, Business creation. Screens: Landing, Signup, Login, empty Dashboard shell. Database: User, Organization, Business, BusinessProfile (empty). APIs: /auth/*, /business (CRUD). AI: none yet. Tests: auth flow, tenant isolation. Definition of Done: a second business created by a second user cannot see the first business's data. Expected User Outcome: "I have an account and a workspace." Expected Business Outcome: foundation for every later milestone.

MILESTONE 2 — Business Onboarding + AI *Outcome:* A user describes their idea and receives an AI-drafted business profile they can review and approve. Features: idea capture flow, Business Setup Agent, Advisor Agent v1, Dashboard v1. Screens: Idea capture, AI recommendation review, Dashboard (health cards + recommendations). Database: AIAgentRun, AIConversation, BusinessProfile (populated fields). APIs: /business/analyze, /ai/assistant/message. AI: Business Setup Agent, Advisor Agent (basic). Tests: draft-then-approve flow, approval gating. Definition of Done: idea text in → approved business profile out, end to end. Expected User Outcome: "BusinessOS understood my idea and helped me structure it." Expected Business Outcome: core differentiated value proposition first demonstrated.

MILESTONE 3 — Products + Services *Outcome:* A user creates products/services with AI-generated content. Features: Product CRUD, Product Content Agent. Screens: Products list, Product create/edit with AI panel. Database: Product, ProductVariant (minimal). APIs: /products, /ai/product-content. AI: Product Content Agent. Tests: content draft/accept flow. Definition of Done: a product can be created, AI-drafted, edited, and published. Expected User Outcome: "I don't have to write my own product descriptions." Expected Business Outcome: second core agent live; retention driver.

MILESTONE 4 — CRM + Customers *Outcome:* A user can track customers and basic order history; first Team Member roles introduced. Features: Customer CRUD, TeamMember + Role. Screens: Customers list/detail, basic Settings (team). Database: Customer, Lead, TeamMember, Role. APIs: /customers, /team. AI: none new (CRM Agent deferred). Tests: role-based access. Definition of Done: a second team member with a restricted role can log in and see only permitted screens. Expected User Outcome: "I know who my customers are." Expected Business Outcome: groundwork for multi-user businesses.

MILESTONE 5 — Sales + Orders *Outcome:* A user can record and track orders against products and customers. Features: Order CRUD, order-customer-product linkage. Screens: Orders list/detail, order entry form. Database: Order, OrderItem. APIs: /orders. AI: none new. Tests: order totals, customer stats update. Definition of Done: an order updates customer history and appears in dashboard metrics. Expected User Outcome: "I can see my sales in one place." Expected Business Outcome: MVP is now functionally complete (Section 19).

MILESTONE 6 — AI Marketing *Outcome:* Marketing Agent and Social Media module introduced. Features: campaign drafting, social scheduling. Screens: Marketing, Campaigns. Database: MarketingCampaign, SocialAccount, SocialPost. APIs: /marketing/*, /social/*. AI: Marketing Agent, Social Media Agent. Tests: draft-review-publish flow, approval gating on publish. Definition of Done: a campaign can be drafted, approved, and scheduled. Expected User Outcome: "AI helps me market, not just describe products." Expected Business Outcome: differentiation vs. pure e-commerce tools sharpens.

MILESTONE 7 — Inventory *Outcome:* Stock tracking and low-stock alerting. Features: Inventory CRUD, Supplier records, Inventory Agent. Screens: Inventory. Database: Inventory, Supplier. APIs: /inventory, /suppliers. AI: Inventory Agent. Tests: stock decrement on order, threshold alerting. Definition of Done: an order correctly decrements stock and triggers an alert at threshold. Expected User Outcome: "I won't oversell what I don't have." Expected Business Outcome: operational depth for physical-product businesses.

MILESTONE 8 — Finance *Outcome:* Invoicing and basic transaction tracking. Features: Invoice CRUD, Transaction records, Finance Assistant Agent. Screens: Finance. Database: Invoice, Transaction. APIs: /finance/*. AI: Finance Assistant Agent. Tests: invoice-to-transaction reconciliation. Definition of Done: an invoice can be drafted, sent (approval-gated), and marked paid with a linked transaction. Expected User Outcome: "I have basic financial visibility without separate accounting software." Expected Business Outcome: reduces reliance on external finance tools.

MILESTONE 9 — Automation *Outcome:* Configurable trigger→action automation rules. Features: Automation rule builder, Automation Agent, event bus/workflow engine. Screens: Automations. Database: Automation. APIs: /automations. AI: Automation Agent. Tests: end-to-end trigger firing, approval gating enforced. Definition of Done: a user-configured rule fires correctly and logs to audit. Expected User Outcome: "Repetitive tasks now happen without me." Expected Business Outcome: time-saved metric (Section 24) becomes measurable.

MILESTONE 10 — Analytics *Outcome:* Cross-module insight generation. Features: Analytics aggregation, Analytics Agent narratives. Screens: Analytics. Database: none new (derived views). APIs: /analytics/*. AI: Analytics Agent. Tests: metric accuracy vs. raw data. Definition of Done: dashboard and Analytics agree on every number. Expected User Outcome: "I understand *why*, not just *what*." Expected Business Outcome: supports upsell into higher tiers (Section 22).

MILESTONE 11 — Storefront *Outcome:* Public-facing website/store per business. Features: template selection, branding, public page publishing, customer-originated orders. Screens: Website builder, public Storefront. Database: Website, Storefront. APIs: /website/*, public storefront APIs. AI: none new (reuses Product Content Agent output). Tests: public order correctly creates an Order/Customer in the owner's workspace. Definition of Done: an external customer can complete a purchase that appears correctly inside BusinessOS. Expected User Outcome: "My whole online presence lives in one system." Expected Business Outcome: unlocks the Phase 10 marketplace path.

MILESTONE 12 — Scale + Advanced AI *Outcome:* Growth Advisor Agent, full agent memory/RAG, multi-provider model routing, and marketplace groundwork. Features: advanced agent roster, memory/RAG infrastructure, model routing, early marketplace discovery. Screens: expanded AI Assistant, marketplace discovery (early). Database: AI_MEMORY, embeddings, marketplace-supporting tables. APIs: expanded /ai/* surface. AI: full roster from Section 9. Tests: cross-agent orchestration, cost/latency benchmarks for model routing. Definition of Done: agents share context appropriately without cross-tenant leakage. Expected User Outcome: "BusinessOS actively helps me grow, not just operate." Expected Business Outcome: platform is ready for Phase 10 marketplace expansion.

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49. Most Important Rule

No coding begins after this document. This Master Product Specification — Parts 1 and 2 together — is the first deliverable. It should leave any reader (founder, engineer, AI coding agent, or investor) able to answer: what we are building, why, who uses it, how they use it, what the user sees, what the AI does, what the backend does, what data is stored, what automations run, what the MVP is, what comes later, how the complete system connects, what the finished product looks like, and exactly what gets built first.

Only after this document is reviewed and approved should Milestone 1 (Section 48) begin.

End of Master Product Specification (Parts 1 & 2). No implementation, coding, or Milestone 1 work has been started, per the stated requirement. This document is intended to be the shared reference for product, engineering, and any AI coding agent involved in building BusinessOS AI.